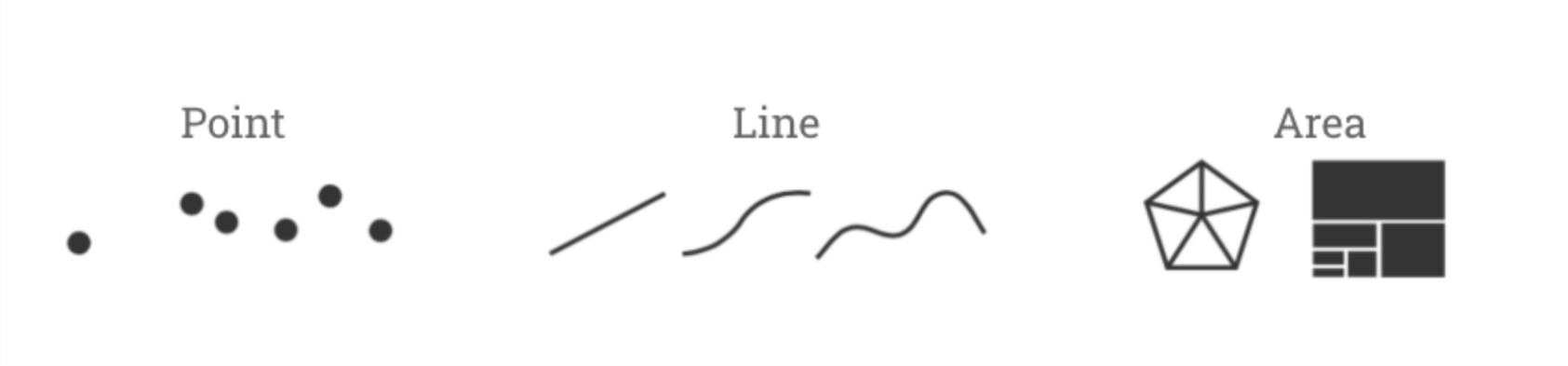
Language of Data Visualization

Marks and Channels



Marks ・　マーク

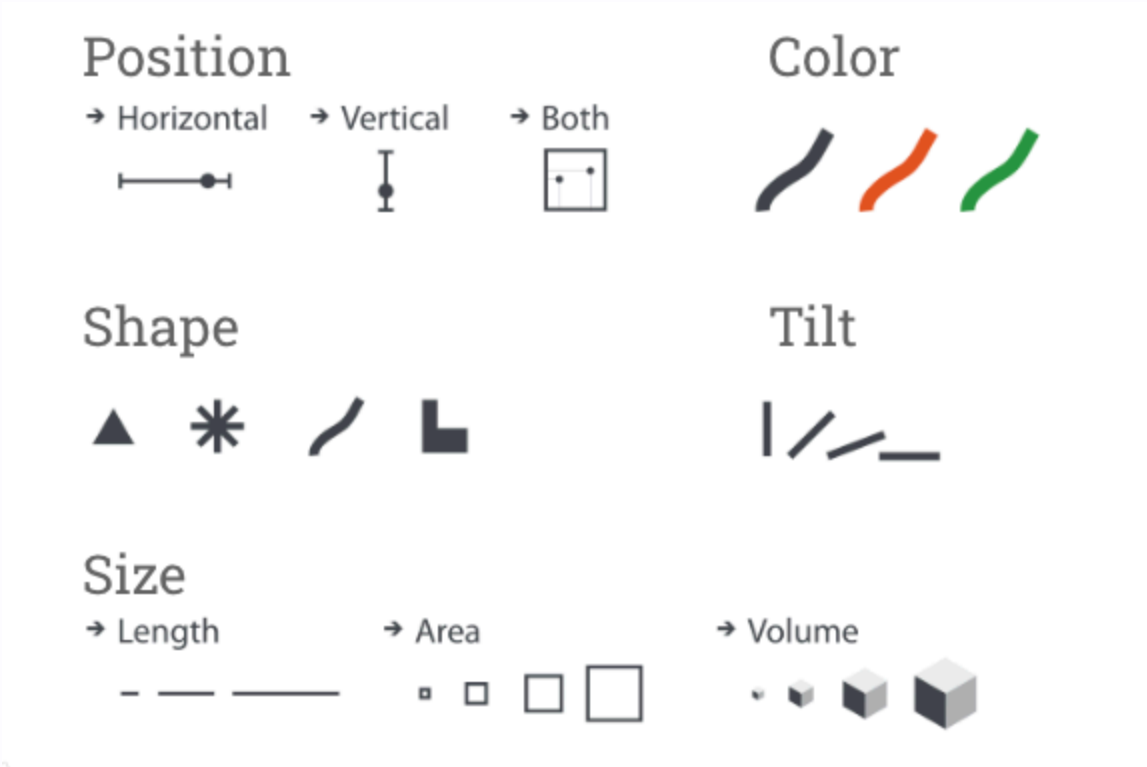
The shapes that represent data.
データを表す形



Graphics from Munzner. Visualization Analysis and Design (2015)

Channels ・ チャンネル

Ways to vary the appearance of marks
マークの見た目を変える方法



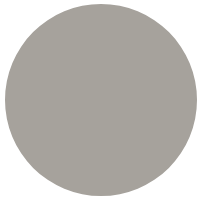
What does a circle represent?

Gapminder: one country, one year



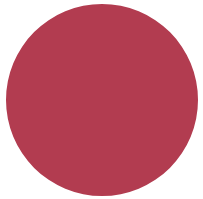
Circle

A mark



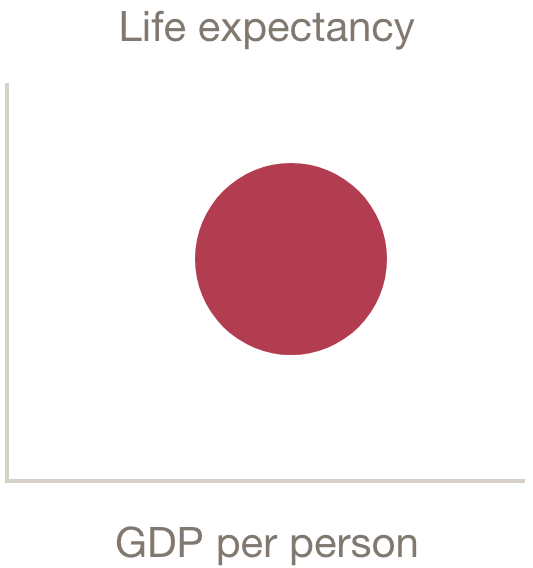
Area

Population



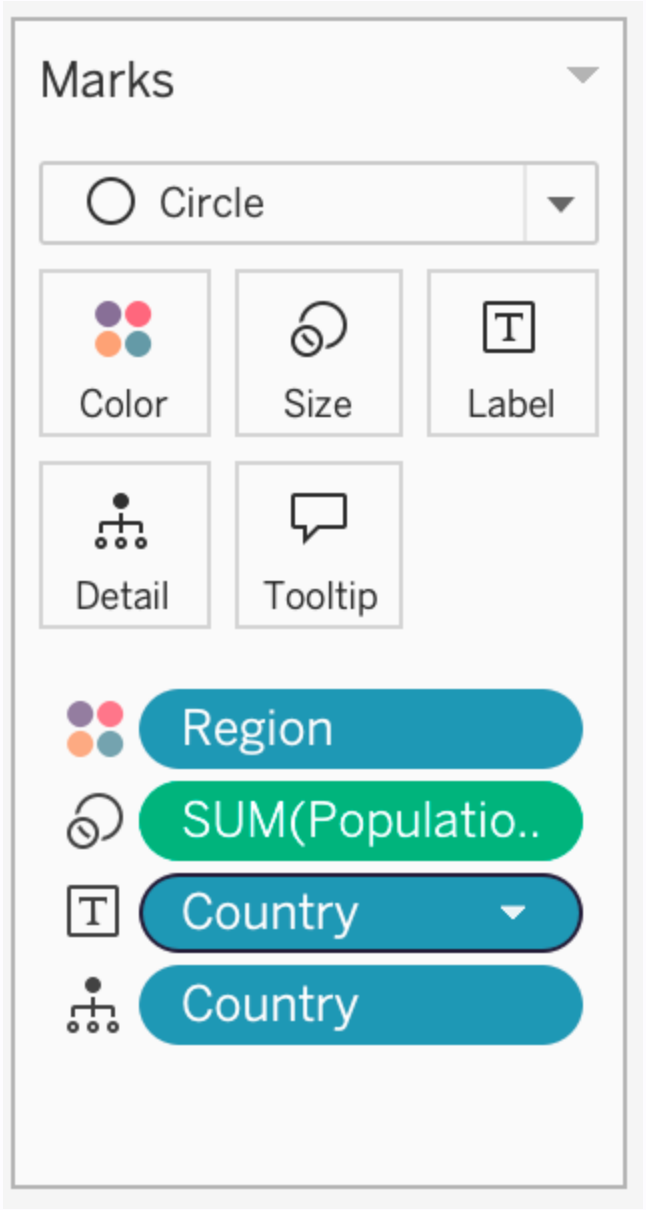
Color

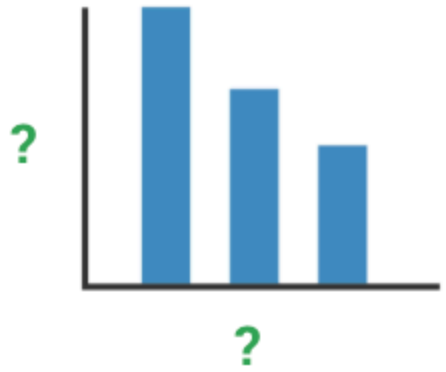
Continent



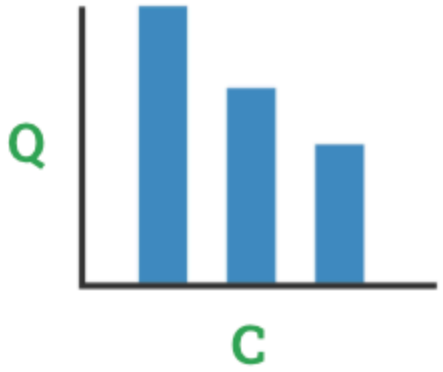
Position

GDP + life expectancy

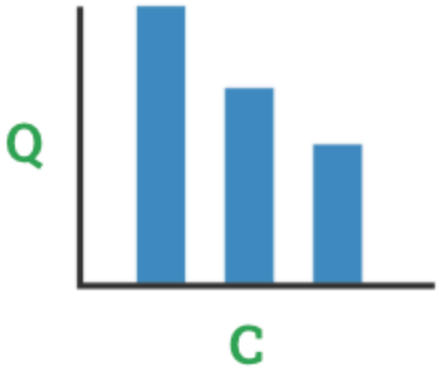

Marks
Circle
Color
Size
Label
Detail
Tooltip
Region
SUM(Populatio..
Country
Country



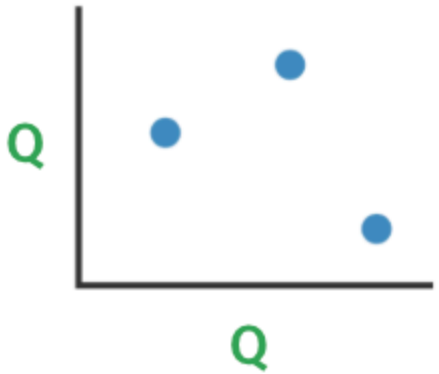
Mark: ?



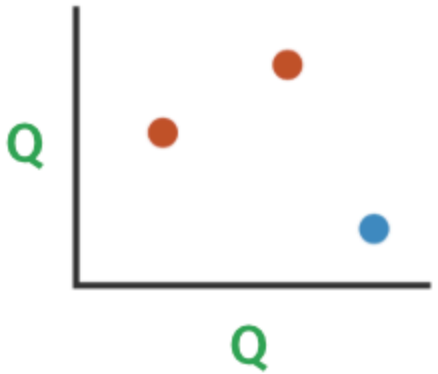
Mark: **line**



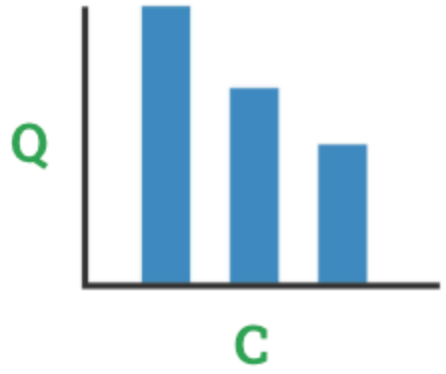
Mark: **line**



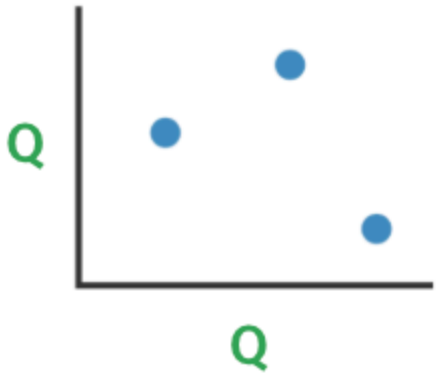
Mark: **point**



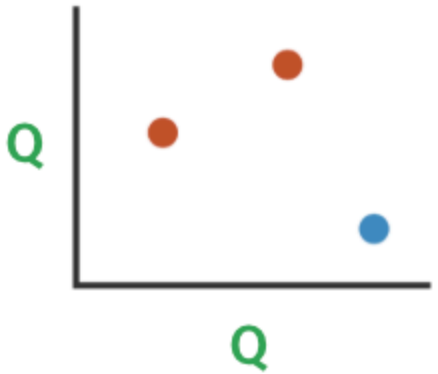
Mark: **point**
Channels:
Color: **C**



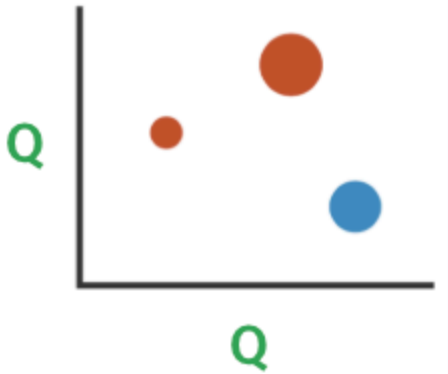
Mark: **line**



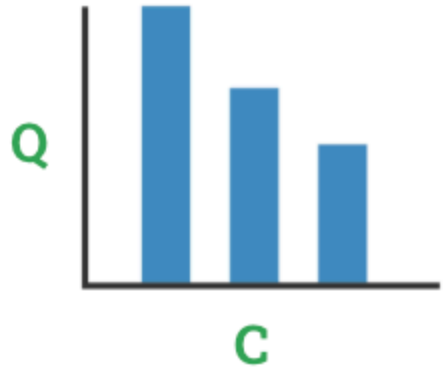
Mark: **point**



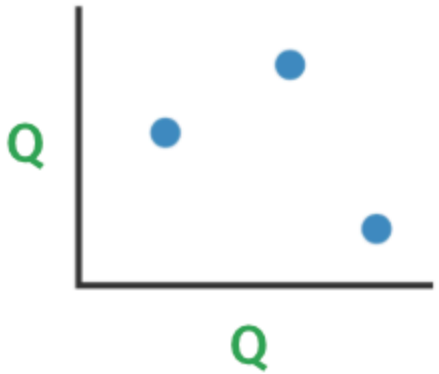
Mark: **point**
Channels:
Color: **C**



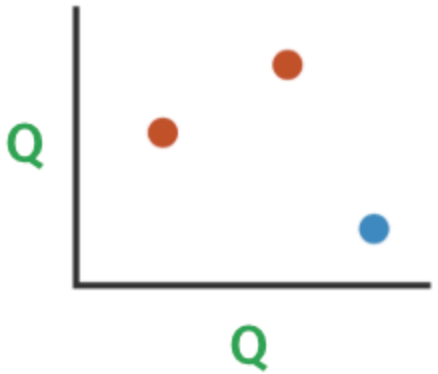
Mark: **point**
Channels: **?**



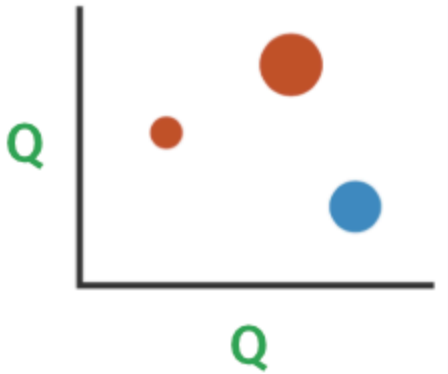
Mark: **line**



Mark: **point**



Mark: **point**
Channels:
Color: **C**



Mark: **point**
Channels:
Color: **C**
Size: **Q**

Position

→ Horizontal → Vertical → Both



Color



Shape



Tilt



Size

→ Length → Area → Volume



Position

→ Horizontal → Vertical → Both



Color



Categorical



Shape



Tilt



×

Ordinal



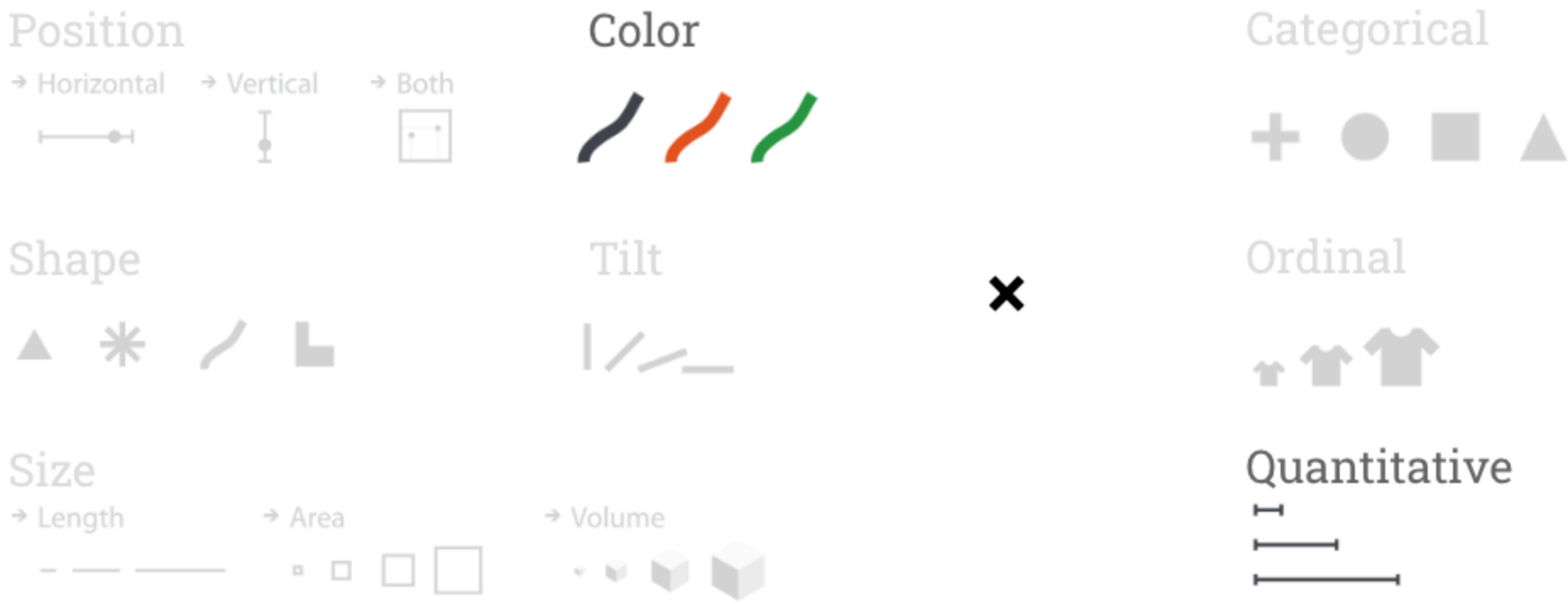
Size

→ Length → Area → Volume



Quantitative




Position
Horizontal
Vertical
Both
Color
Categorical
Shape
Tilt
Ordinal
Size
Length
Area
Volume
Quantitative

Filters

Marks

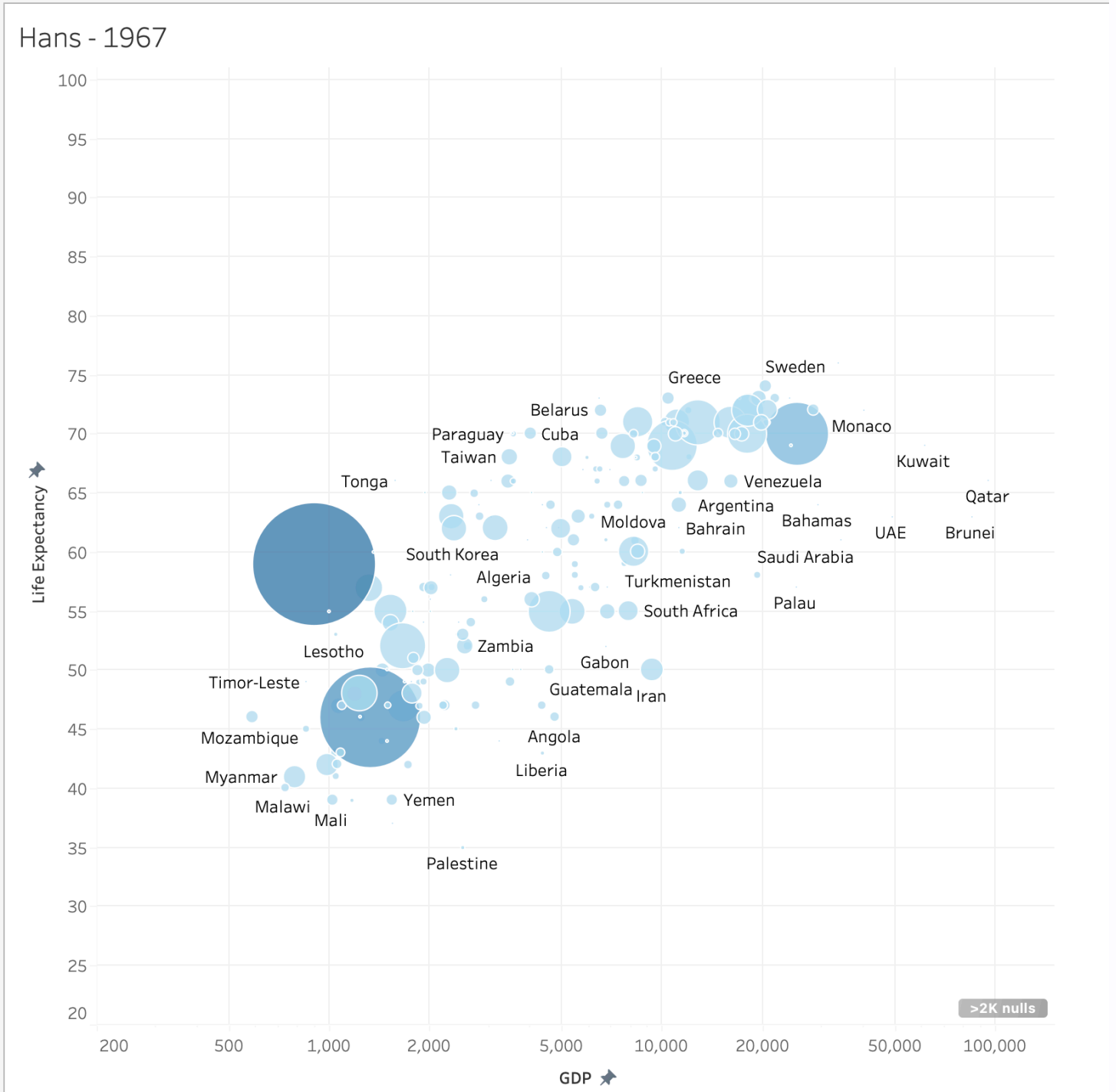
Circle

Color | Size | Label | Detail | Tooltip

SUM(Populatio..
SUM(Populatio..
Country
Country

SUM(Population)

651 999M



Position

→ Horizontal → Vertical → Both

Color



Categorical



Shape

Tilt

×

Ordinal

Size

→ Length → Area → Volume

Quantitative

Filters

Marks

Circle

Color | Size | Label

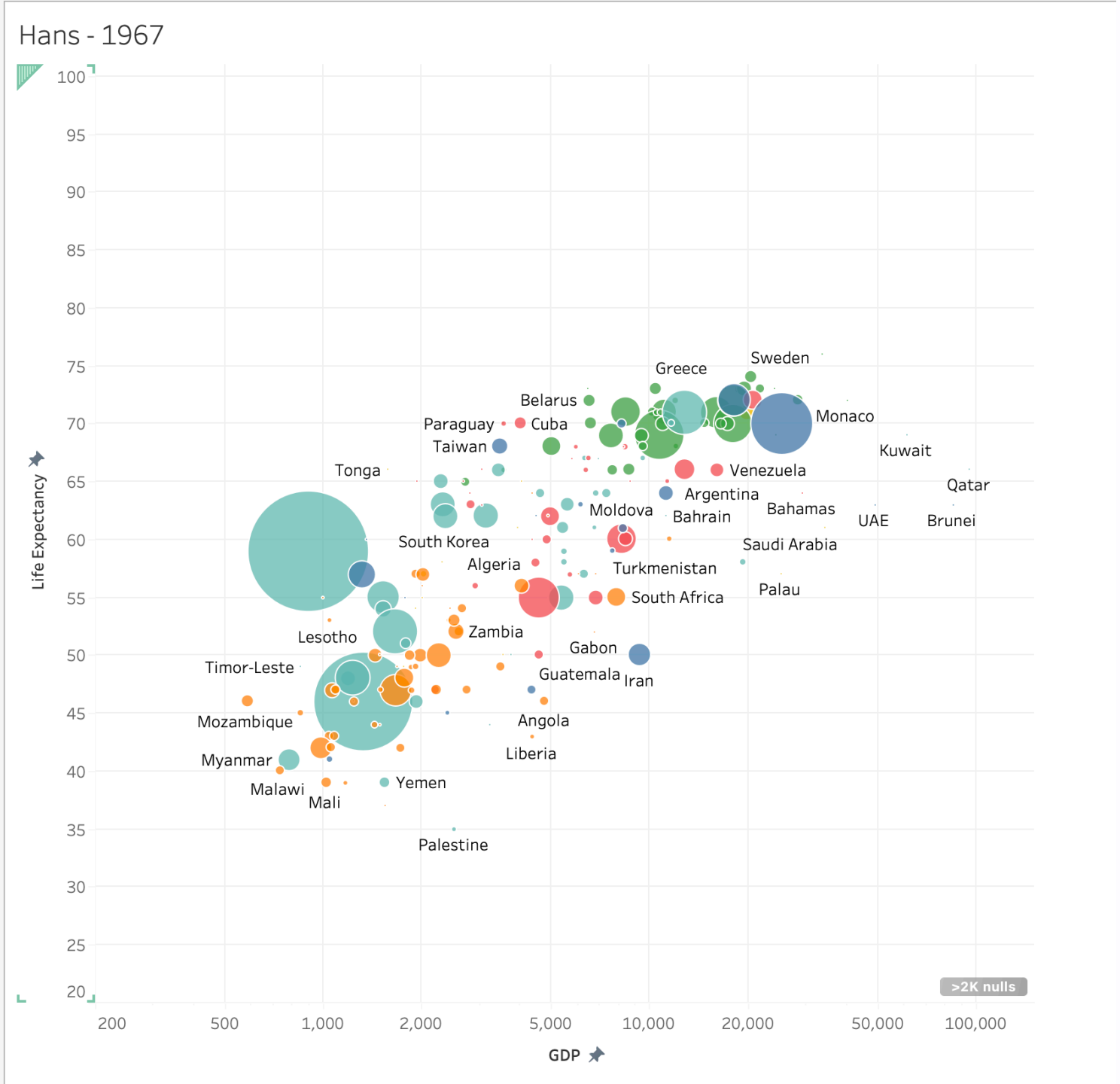
Detail | Tooltip

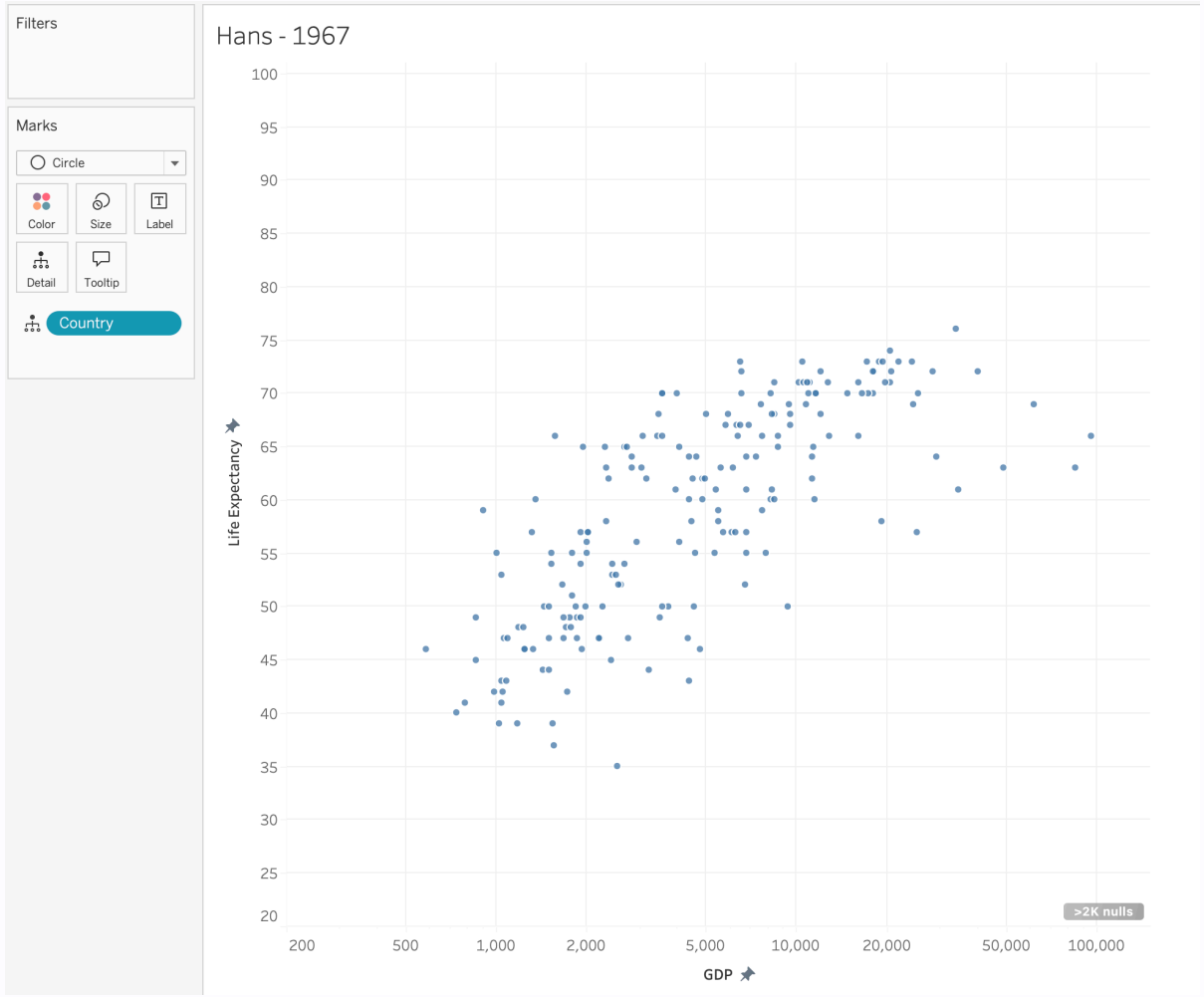
Region

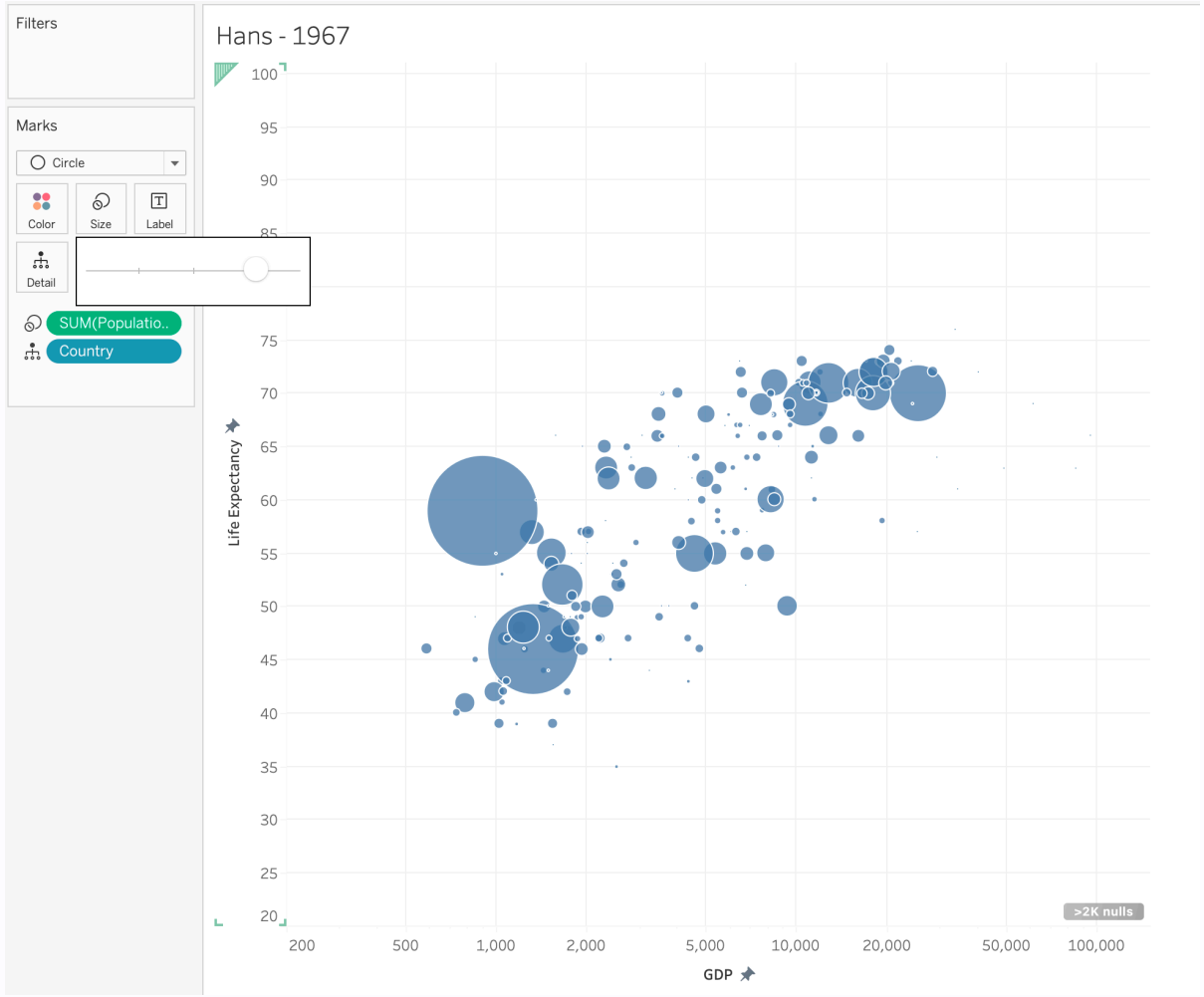
SUM(Populatio..

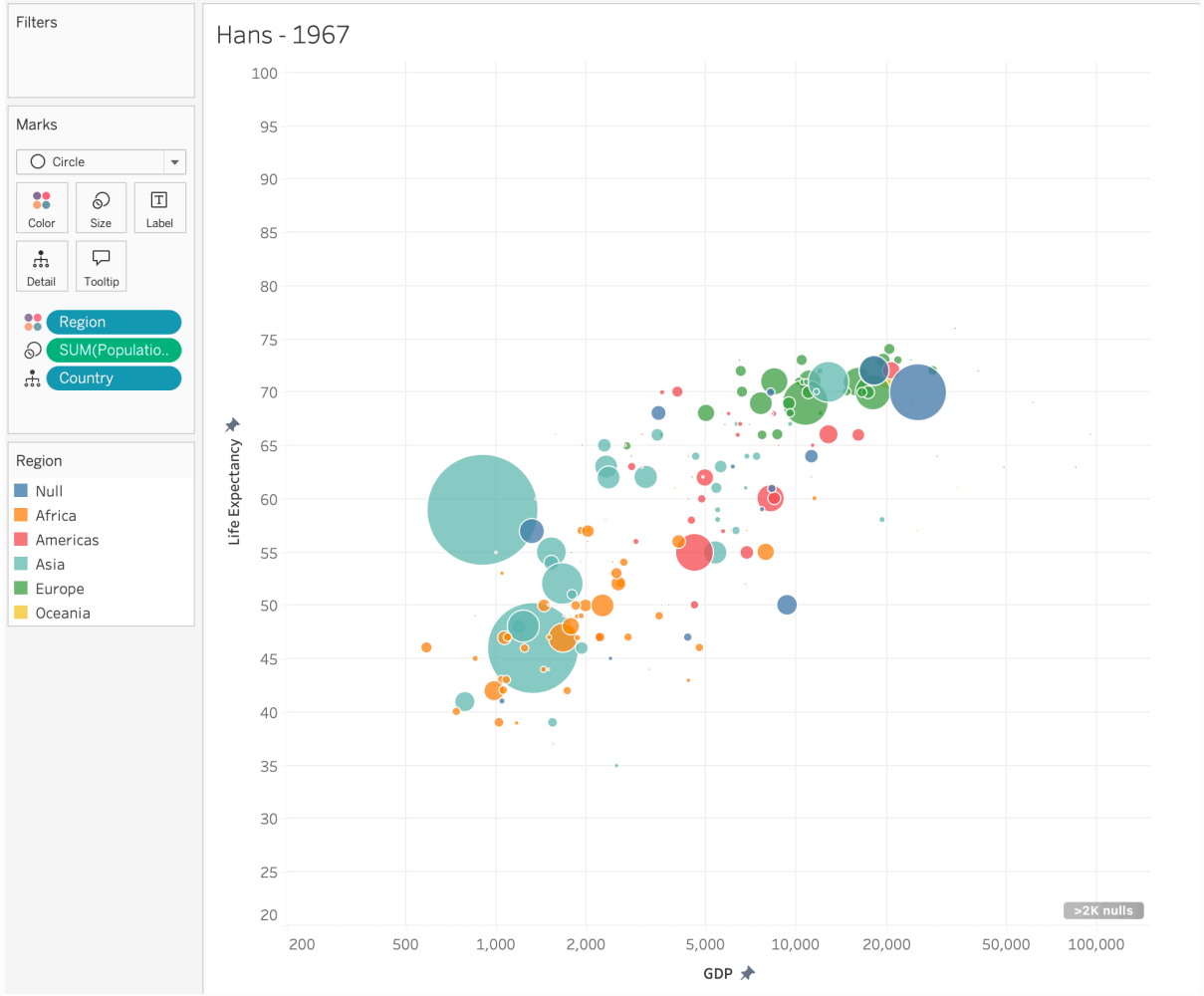
Country

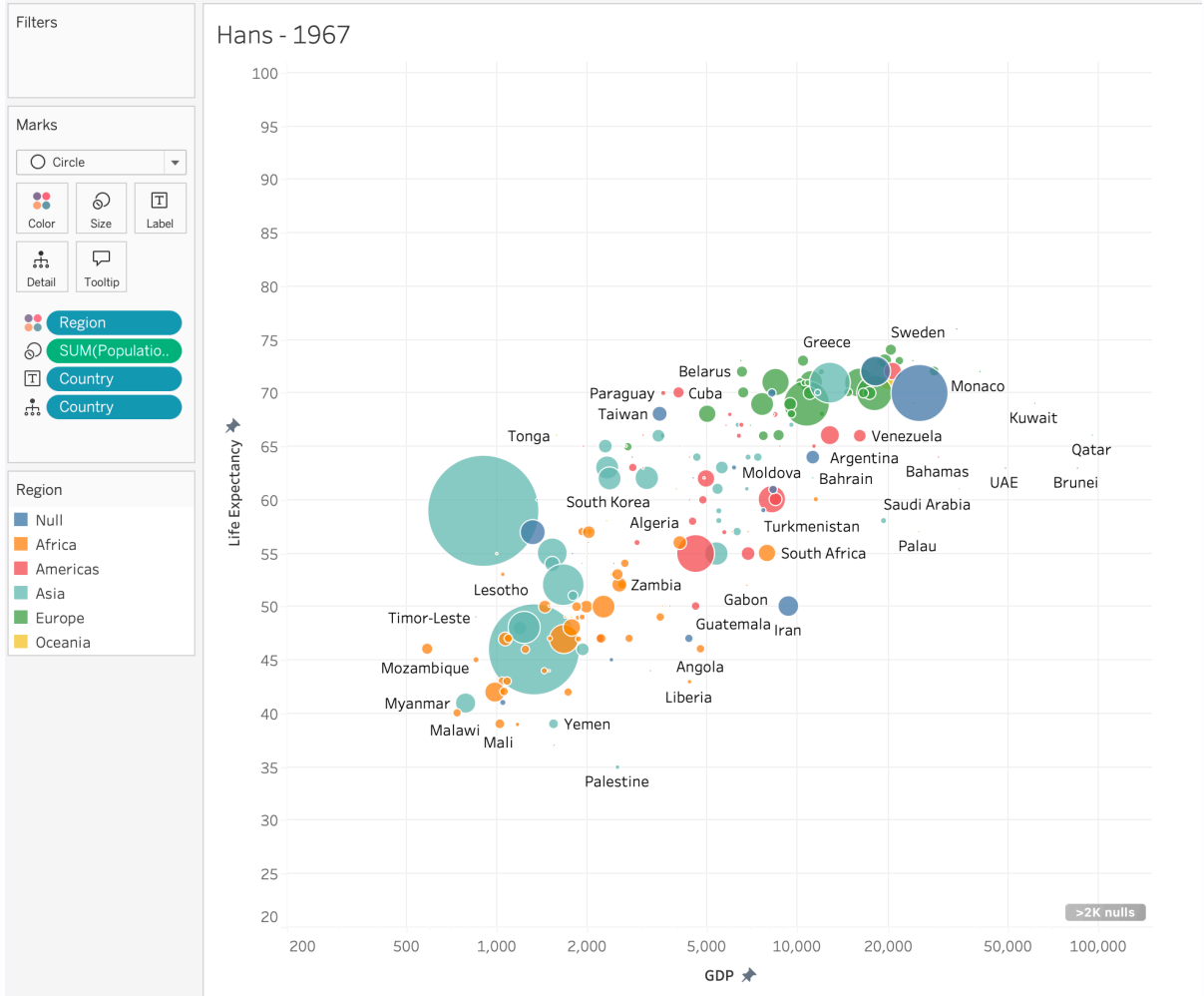
Country

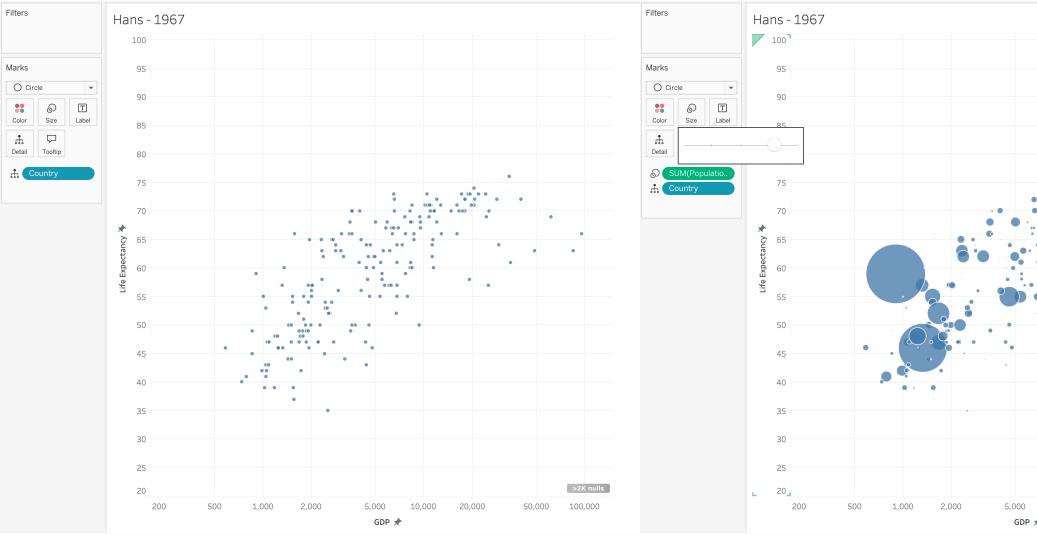


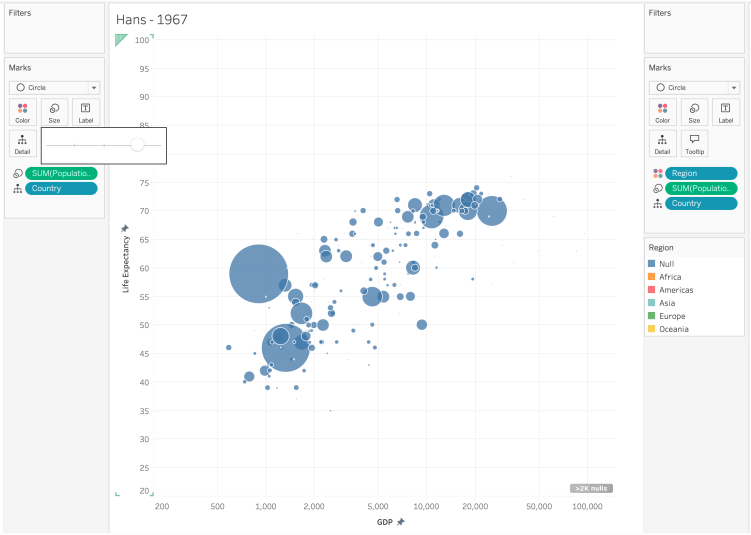

Filters
Marks
Circle
Color
Size
Label
Detail
Tooltip
Country
Hans - 1967
100
95
90
85
80
75
70
65
60
55
50
45
40
35
30
25
20
Life Expectancy
200
500
1,000
2,000
5,000
10,000
20,000
50,000
100,000
GDP
>2K nulls

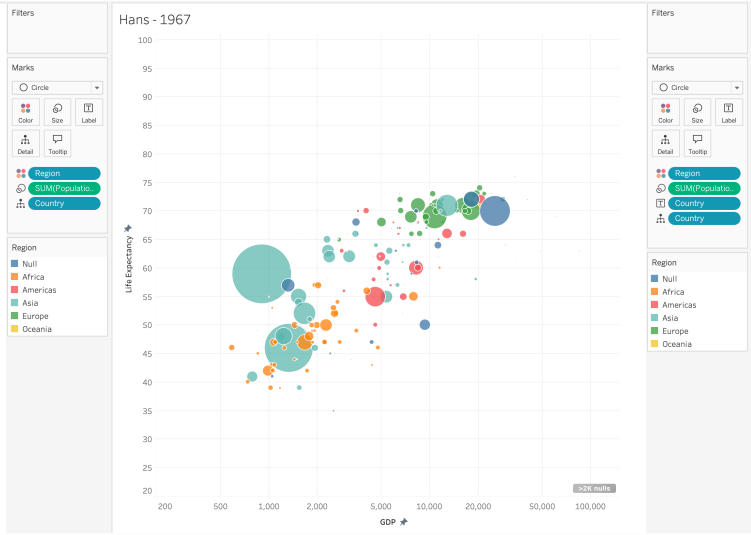

Filters
Marks
Circle
Color
Size
Label
Detail
SUM(Populatio..
Country
Hans - 1967
100
95
90
85
75
70
65
60
55
50
45
40
35
30
25
20
Life Expectancy
200
500
1,000
2,000
5,000
10,000
20,000
50,000
100,000
GDP
>2K nulls


Filters
Marks
Circle
Color
Size
Label
Detail
Tooltip
Region
SUM(Populatio..
Country
Region
Null
Africa
Americas
Asia
Europe
Oceania
Hans - 1967
100
95
90
85
80
75
70
65
60
55
50
45
40
35
30
25
20
Life Expectancy
200
500
1,000
2,000
5,000
10,000
20,000
50,000
100,000
GDP
>2K nulls

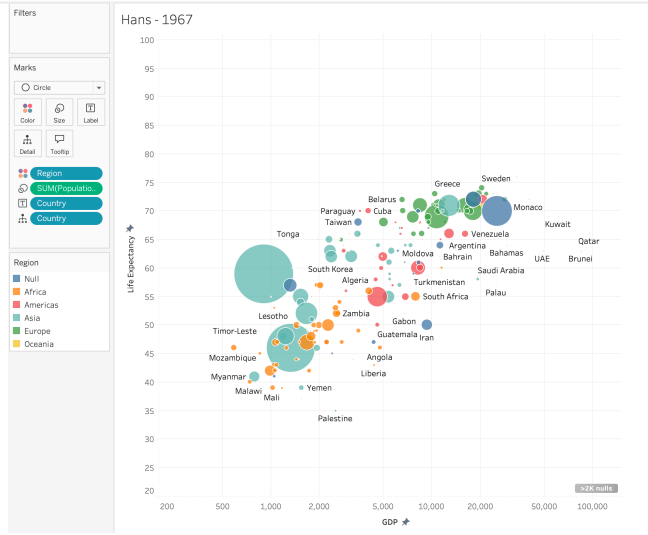

Filters
Marks
Circle
Color
Size
Label
Detail
Tooltip
Region
SUM(Populatio..
Country
Country
Region
Null
Africa
Americas
Asia
Europe
Oceania
Hans - 1967
Life Expectancy
GDP
100
95
90
85
80
75
70
65
60
55
50
45
40
35
30
25
20
200
500
1,000
2,000
5,000
10,000
20,000
50,000
100,000
Greece
Sweden
Belarus
Monaco
Paraguay
Cuba
Taiwan
Kuwait
Tonga
Venezuela
Qatar
Argentina
Moldova
Bahrain
Bahamas
UAE
Brunei
South Korea
Saudi Arabia
Algeria
Turkmenistan
Palau
South Africa
Zambia
Lesotho
Gabon
Timor-Leste
Guatemala
Iran
Mozambique
Angola
Liberia
Myanmar
Malawi
Yemen
Mali
Palestine
>2K nulls


Filters
Marks
Circle
Color
Size
Label
Detail
Tooltip
Country
Hans - 1967
Life Expectancy
GDP
>2K nulls


Filters
Marks
Circle
Color
Size
Label
Detail
SUM(Populatio..
Country
Hans - 1967
Life Expectancy
GDP
>2K nulls


Filters
Marks
Circle
Color
Size
Label
Detail
Tooltip
Region
SUM(Populatio..
Country
Region
Null
Africa
Americas
Asia
Europe
Oceania
Hans - 1967
Life Expectancy
GDP
>2K nulls

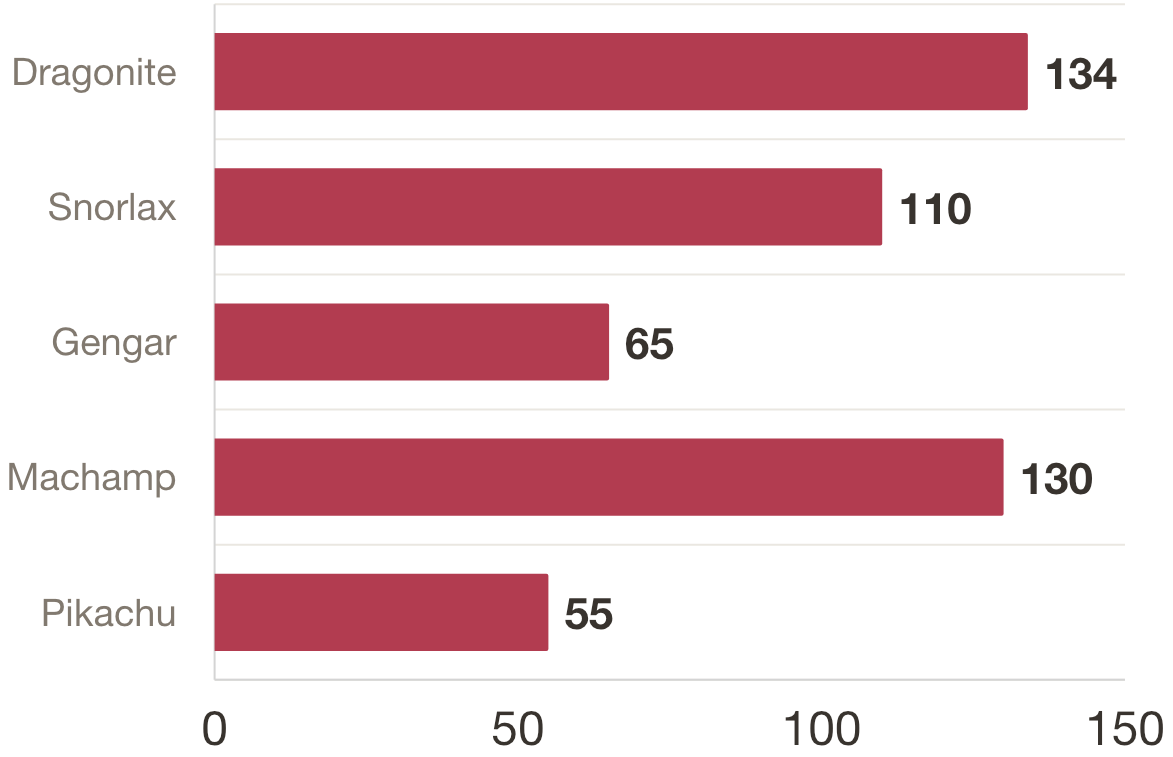
Filters
Marks
Circle
Color
Size
Label
Detail
Tooltip
Region
SUM(Populatio..
Country
Country
Region
Null
Africa
Americas
Asia
Europe
Oceania
Hans - 1967
Life Expectancy
GDP
>2K nulls


Hans - 1967
Life Expectancy
GDP
>2K nulls
Greece
Sweden
Belarus
Monaco
Paraguay
Cuba
Taiwan
Kuwait
Tonga
Venezuela
Argentina
Qatar
Moldova
Bahrain
Bahamas
UAE
Brunei
South Korea
Saudi Arabia
Algeria
Turkmenistan
Palau
South Africa
Lesotho
Zambia
Gabon
Timor-Leste
Guatemala
Iran
Mozambique
Angola
Liberia
Myanmar
Malawi
Yemen
Mali
Palestine

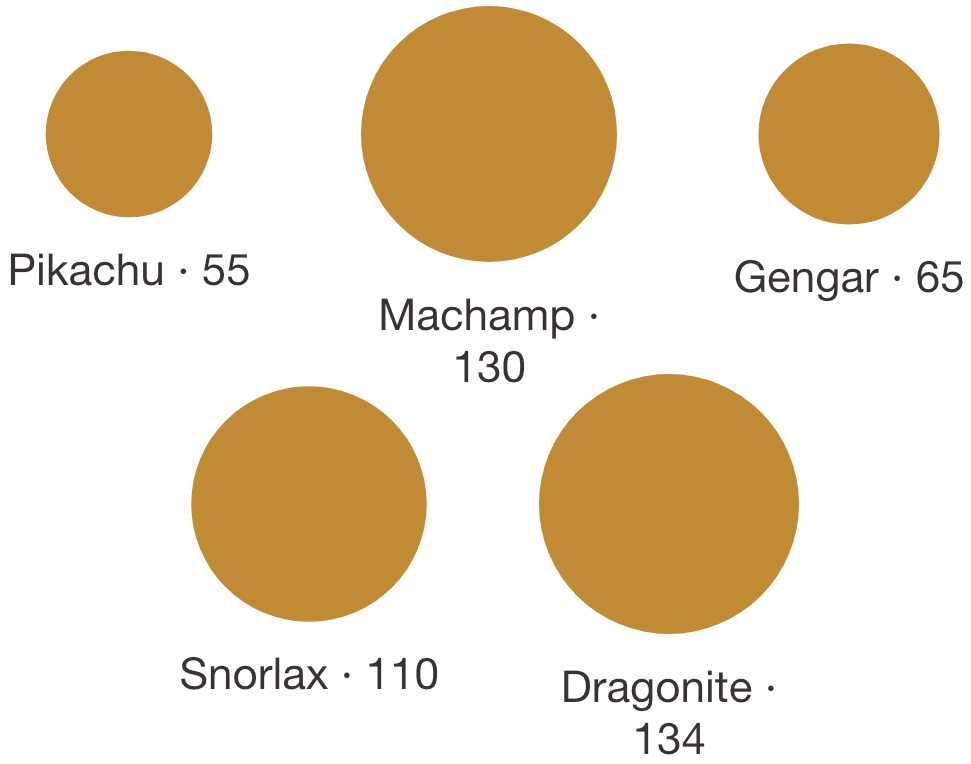
Which comparison is easier?

Same Pokémon. Same Attack values.

POSITION + LENGTH



AREA

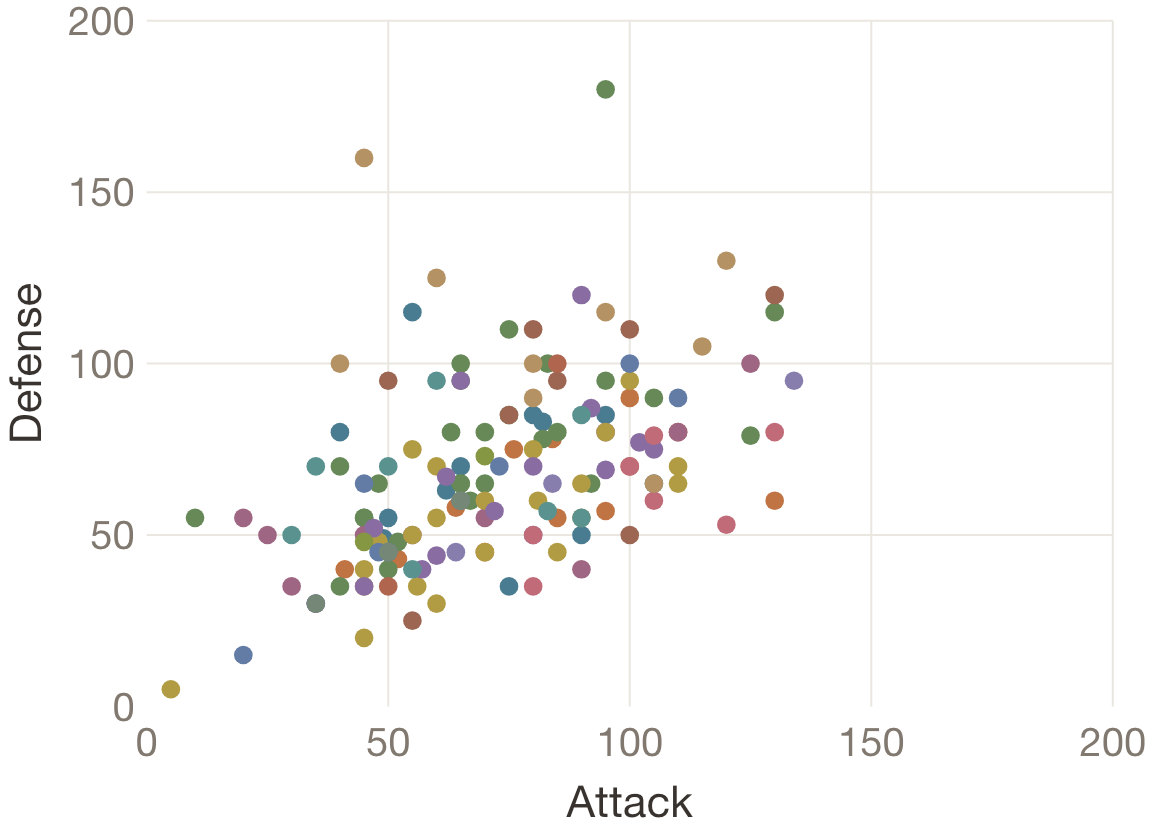


5 selected species · base Attack

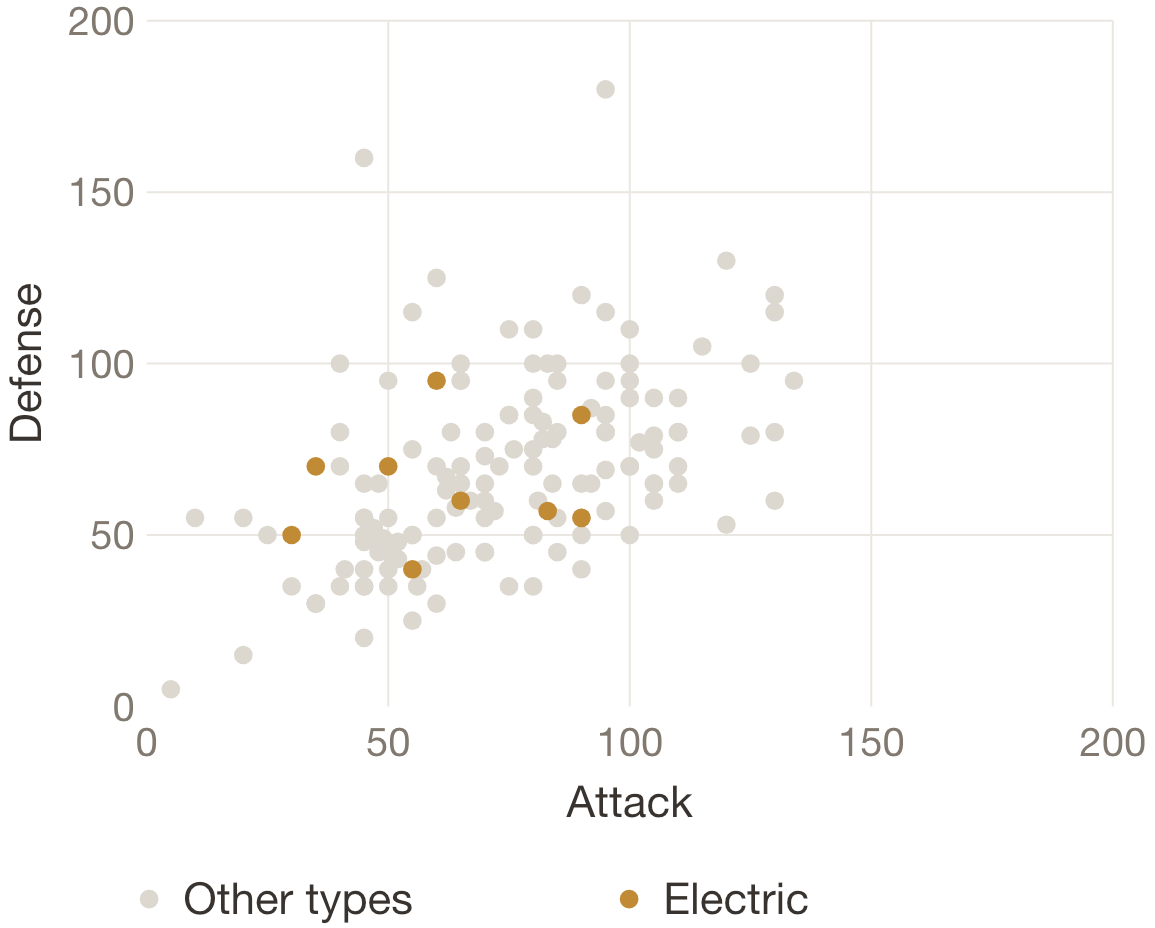
What catches your eye first?

Color everything — or direct attention?

ALL TYPES



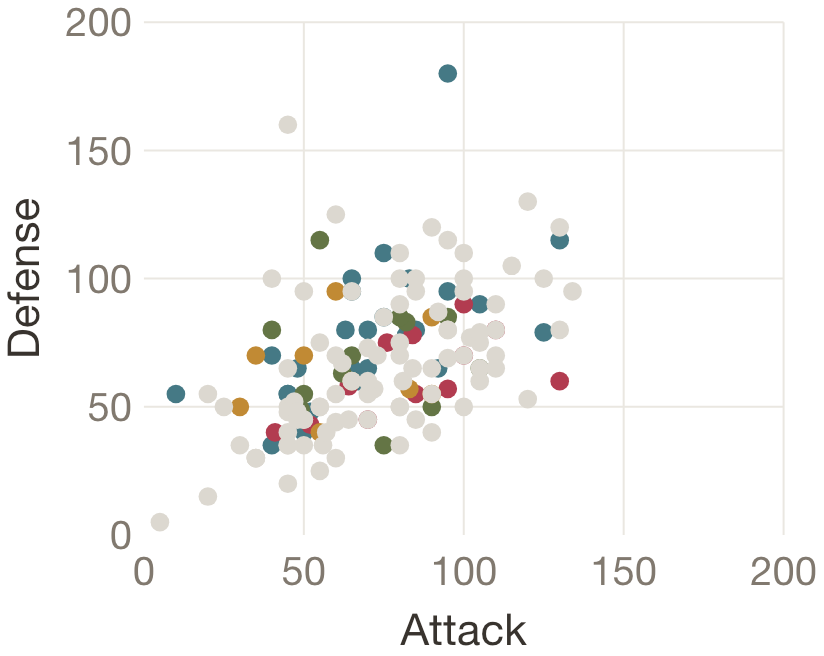
ELECTRIC IN FOCUS



Color needs a job.

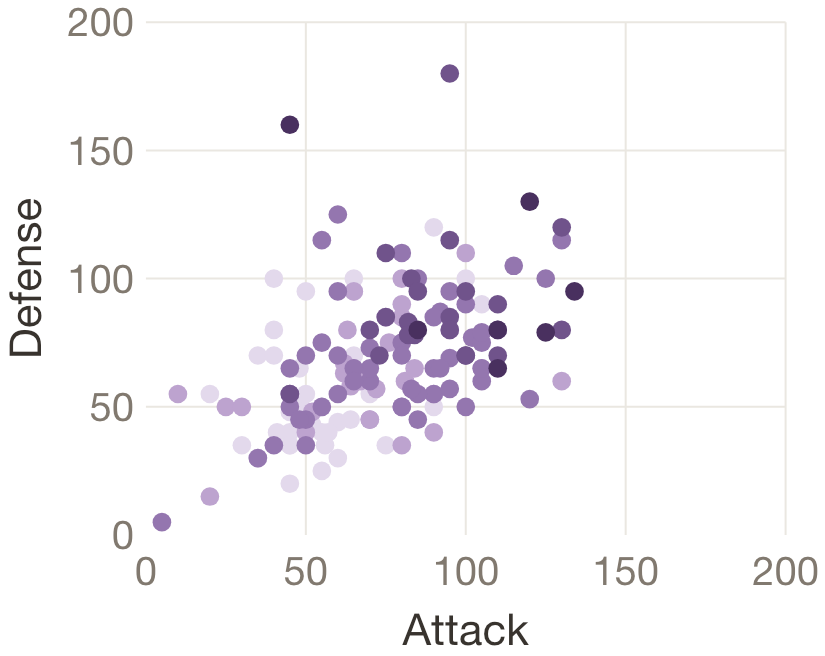
Different kinds of data need different color logic.

TYPE



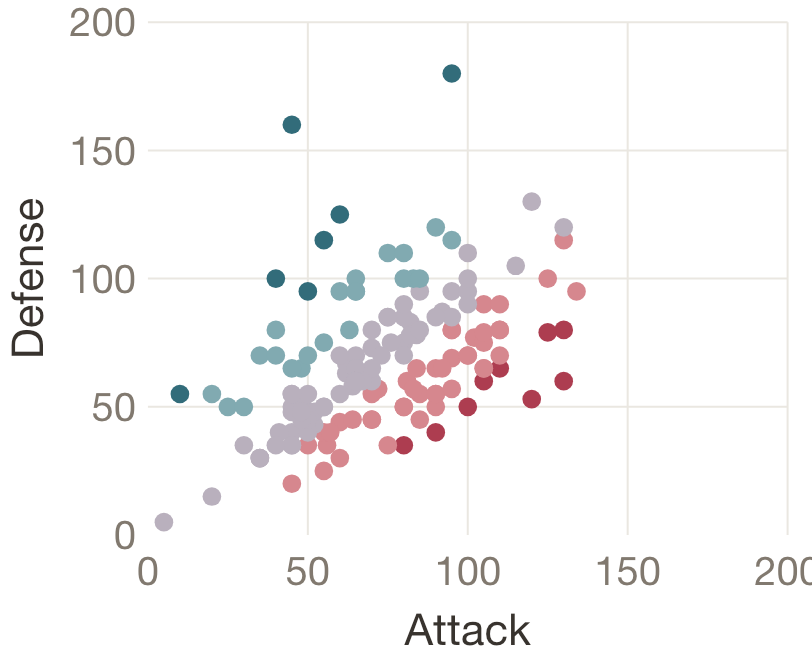
- Water
- Fire
- Grass
- Electric
- Other

WEIGHT



- 0–<10 kg
- 10–<30 kg
- 30–<70 kg
- 70–<150 kg
- 150+ kg

DEFENSE – ATTACK



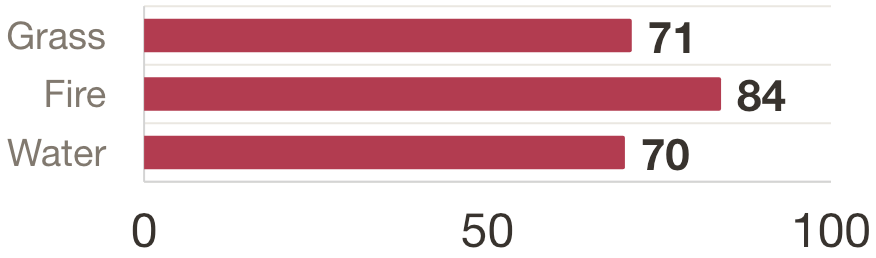
- ≤–41
- –40…–11
- –10…10
- 11…40
- ≥41

Layout creates relationships.

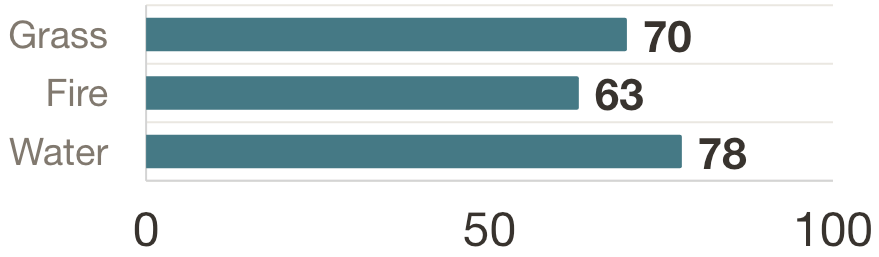
Same values. Different alignment.

SCATTERED

Attack

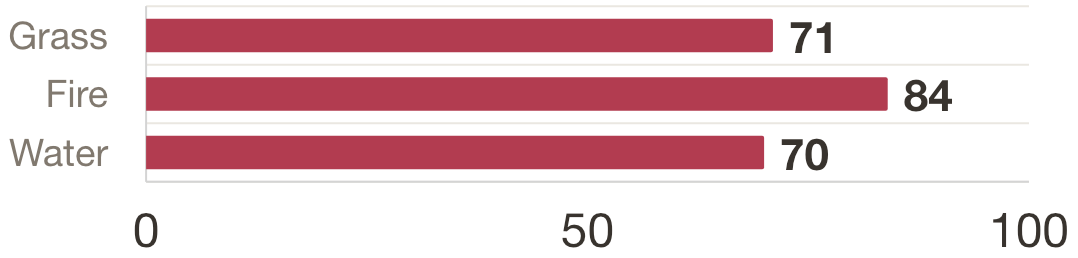


Defense

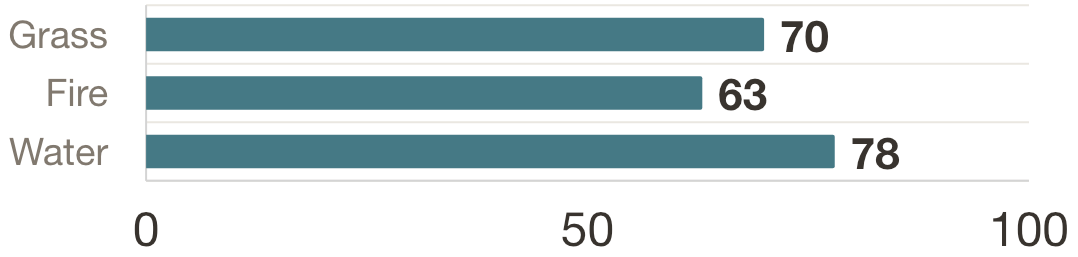


ALIGNED

Attack



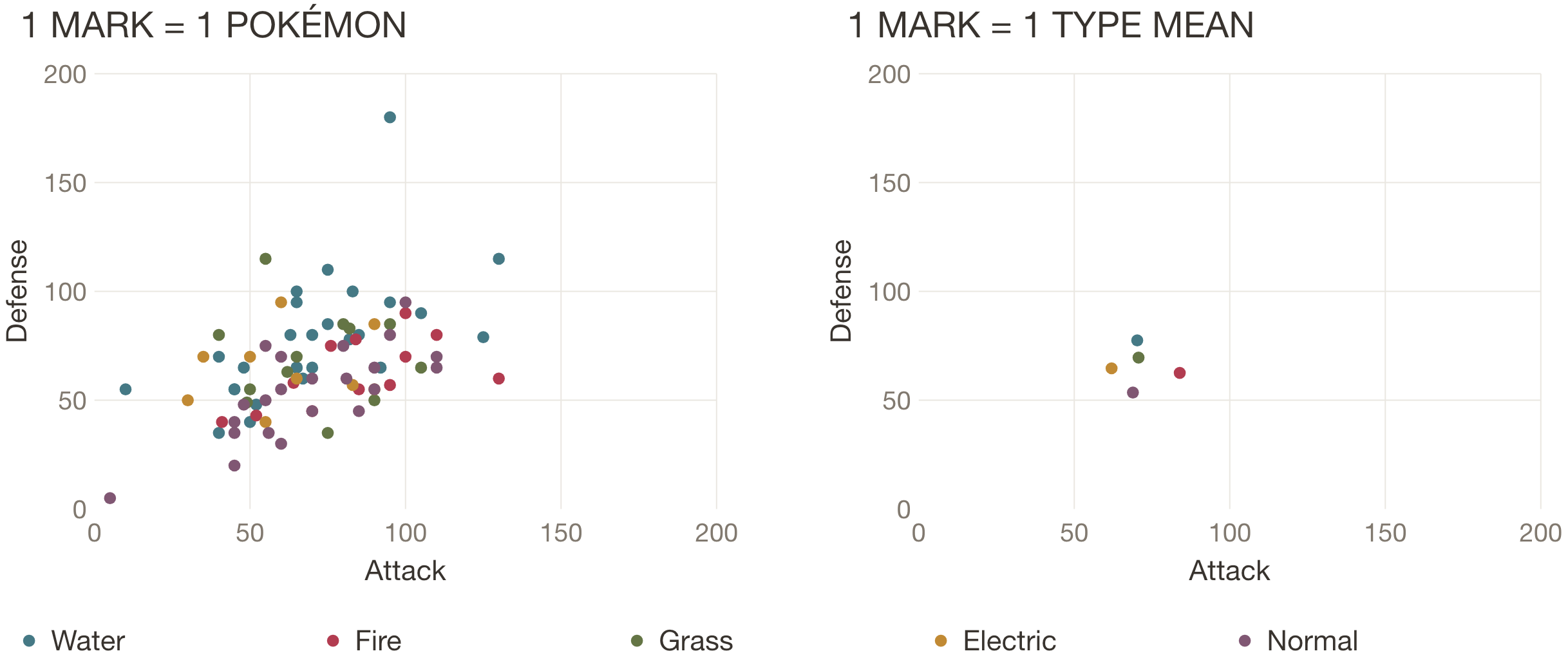
Defense



Mean base stats · primary type · rounded to whole numbers

What does one mark represent?

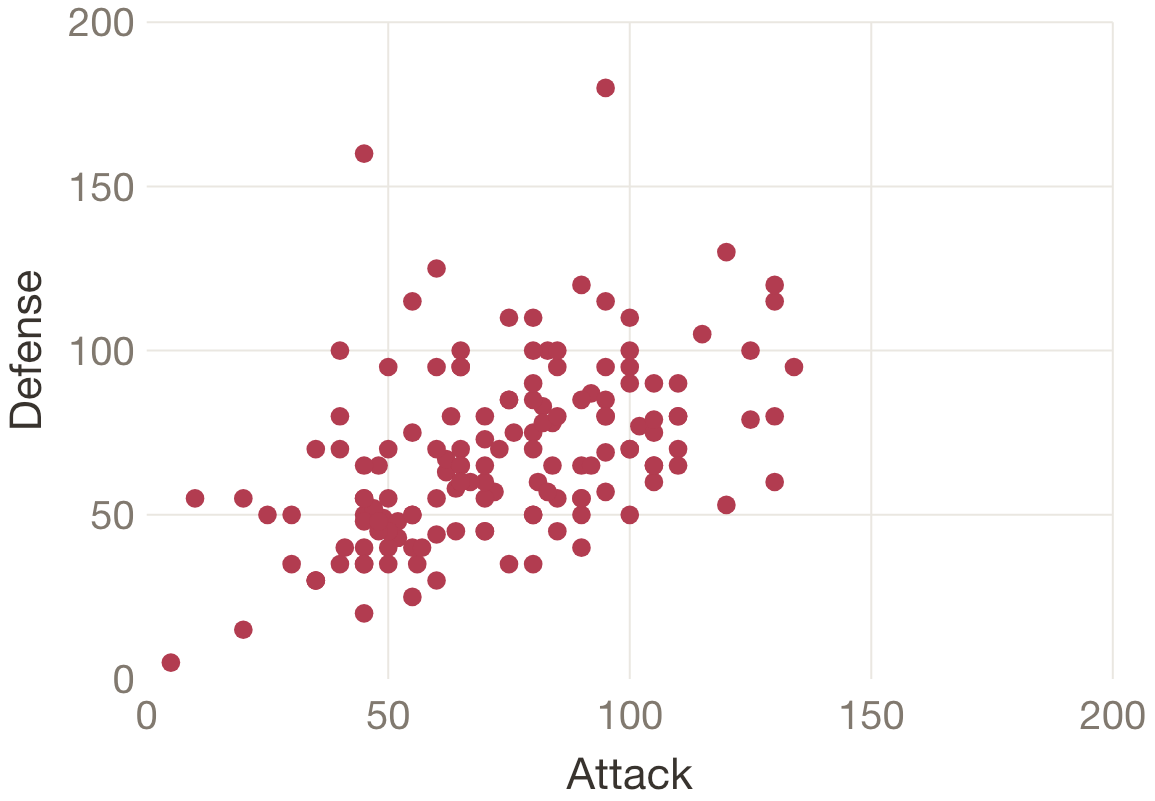
Aggregation changes the story.



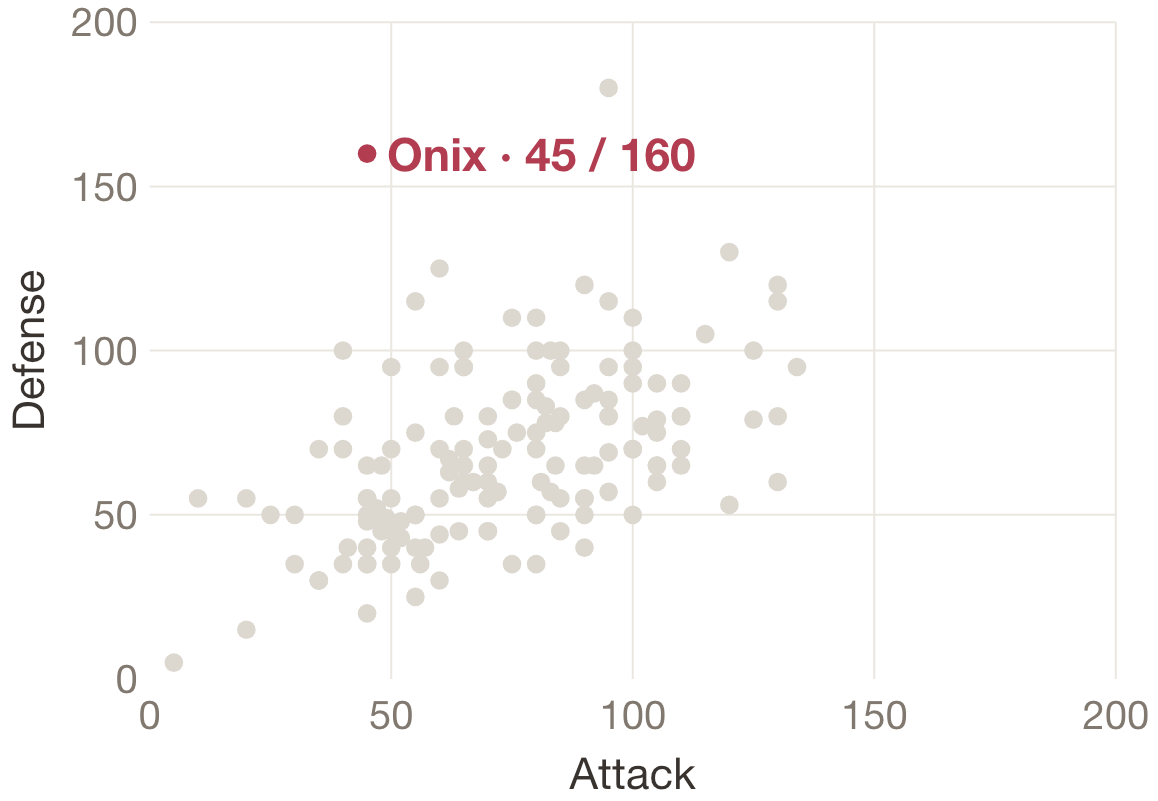
Help the reader find the point.

A title is a chance to say something.

Pokémon stats



Onix: high Defense, low Attack



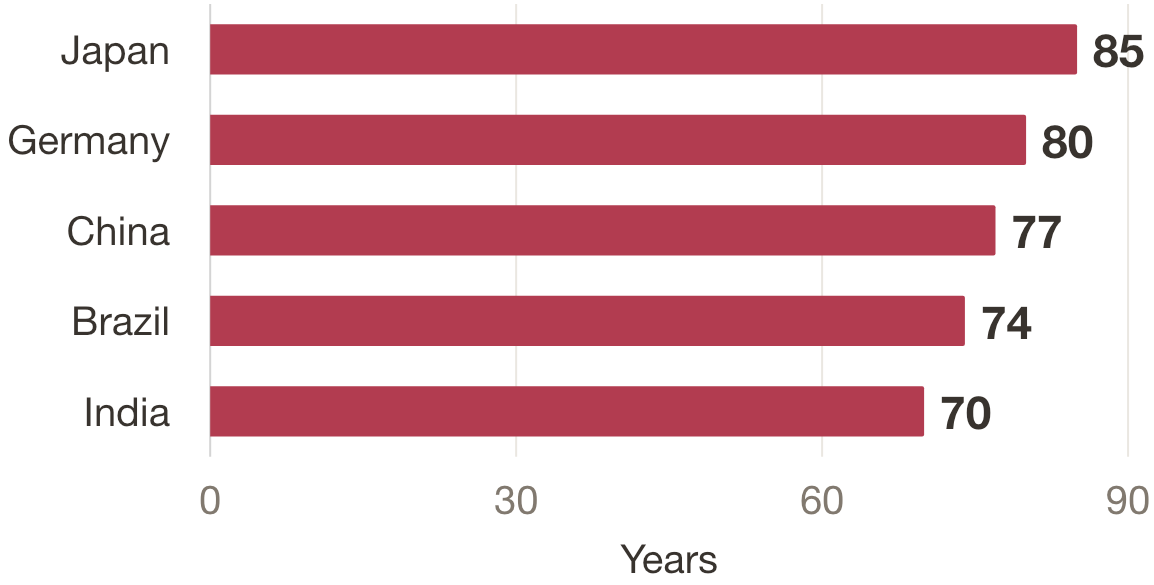
ONE CLEAR CLAIM. ONE USEFUL LABEL.

Tableau chart types: comparisons

Life expectancy in five countries (2020)

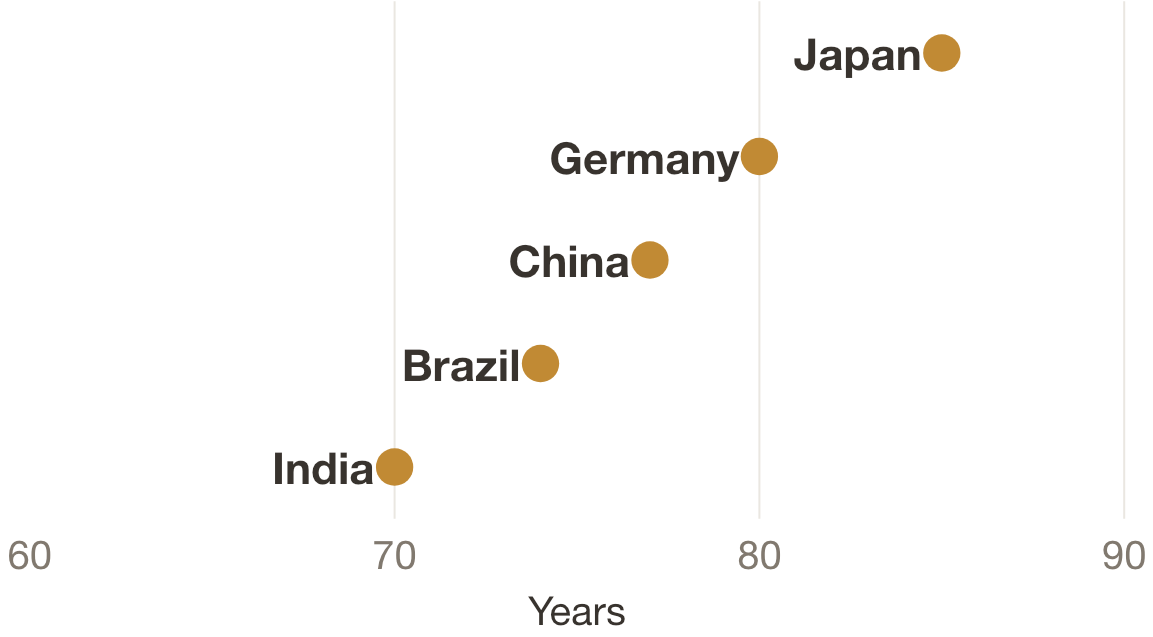
Bar chart

Lengths compare values. Start the scale at zero.



Dot plot

Positions make small differences easier to see.



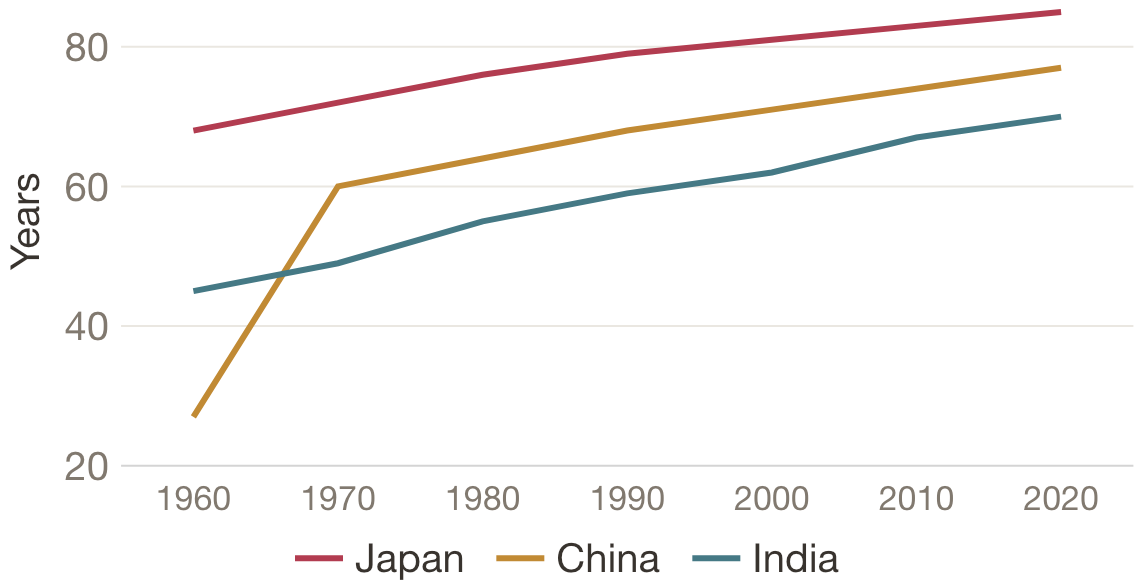
Select fields, then open Show Me. The options depend on your fields.

Tableau chart types: change over time

A line shows a trajectory. An area also emphasizes magnitude.

Line chart

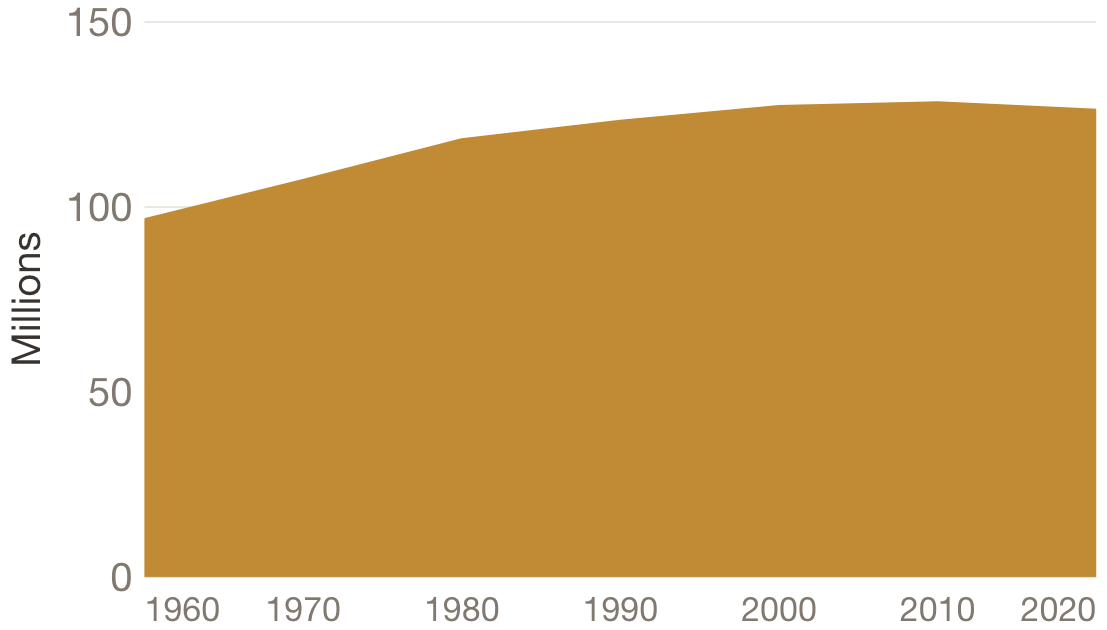
Life expectancy in Japan, China and India



Line: compare trends over time.

Area chart

Japan's population, in millions



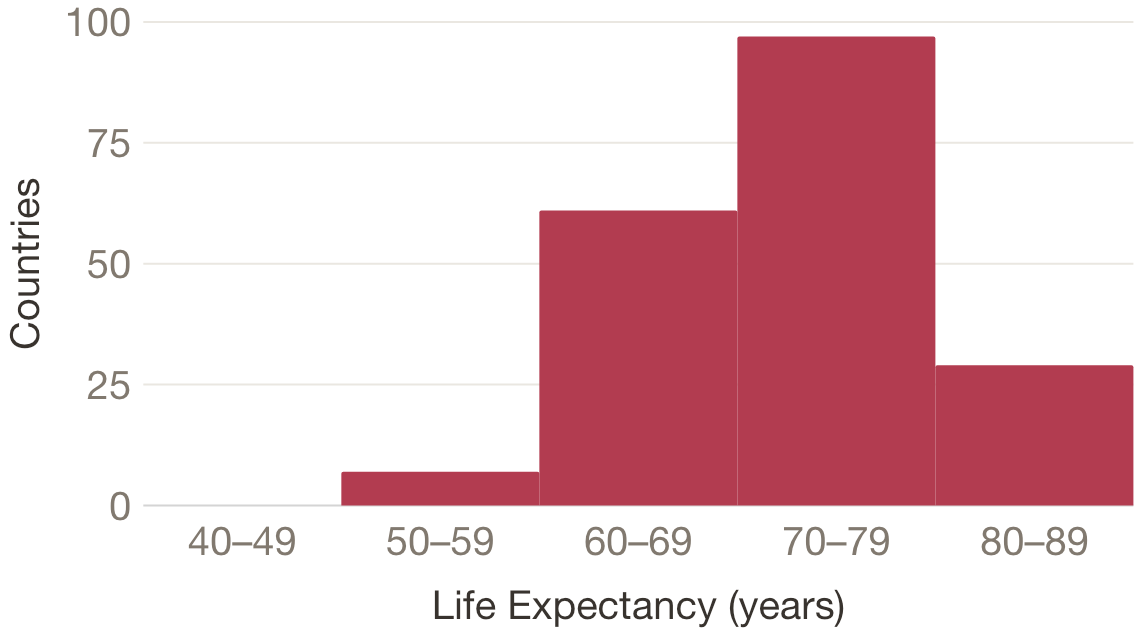
Area: use a meaningful zero baseline.

Tableau chart types: distributions

What is typical, and how much variation is there?

Histogram

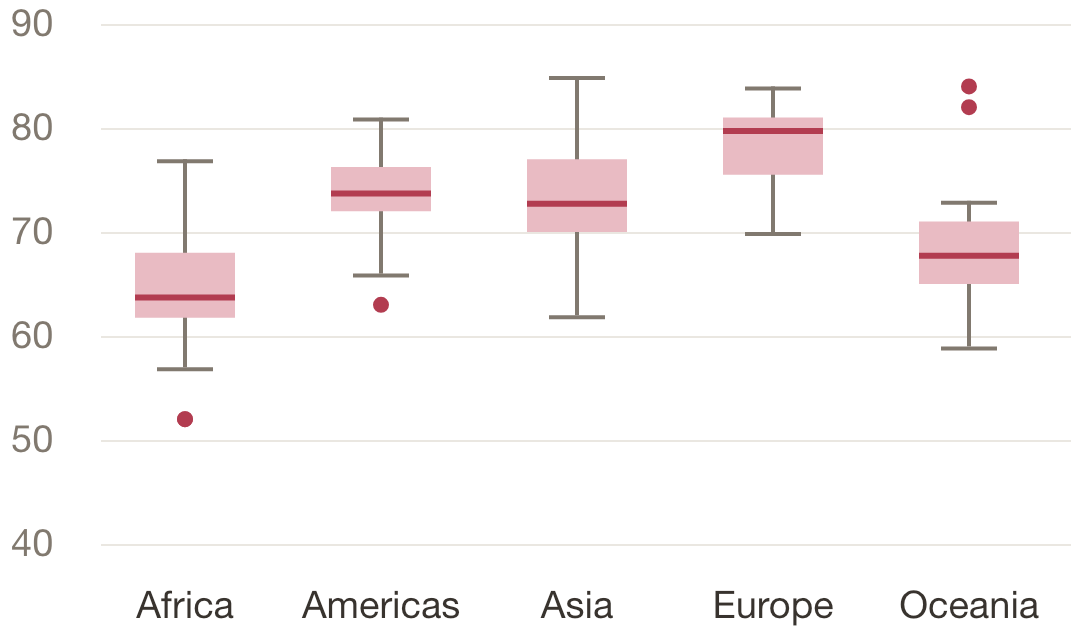
How many countries fall in each range? (2020)



A histogram counts values in numeric bins.

Box-and-whisker plot

Country life expectancy within each region (2020)



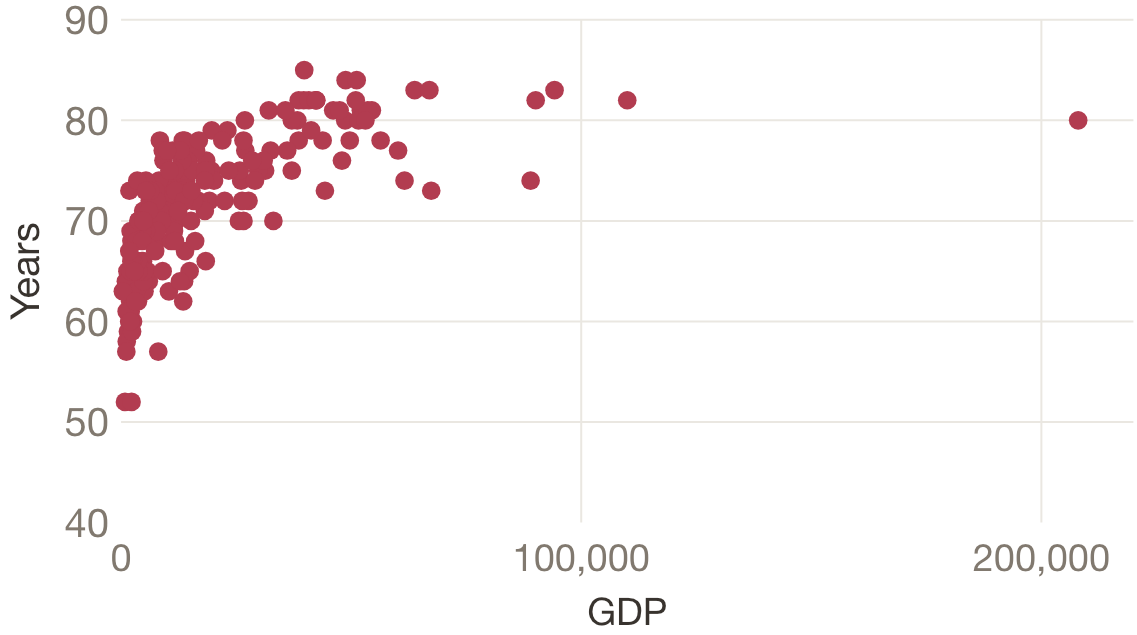
Median, middle 50%, spread and outliers.

Tableau chart types: relationships and patterns

One pair of measures, or a matrix of repeated comparisons

Scatter plot

GDP and life expectancy, one circle per country



Association does not establish cause.

Highlight table

Life expectancy by country and decade

	1980	1990	2000	2010	2020
Japan	76	79	81	83	85
Germany	73	75	78	80	80
China	64	68	71	74	77
Brazil	62	67	71	73	74
India	55	59	62	67	70

40 50 60 70 80 90

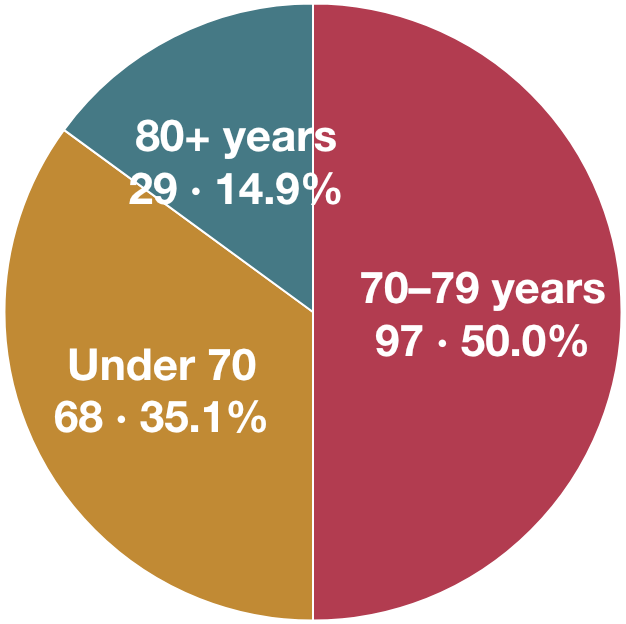
One shared color scale, in years.

Tableau chart types: parts of a whole

How many countries fall in each life-expectancy band? (2020)

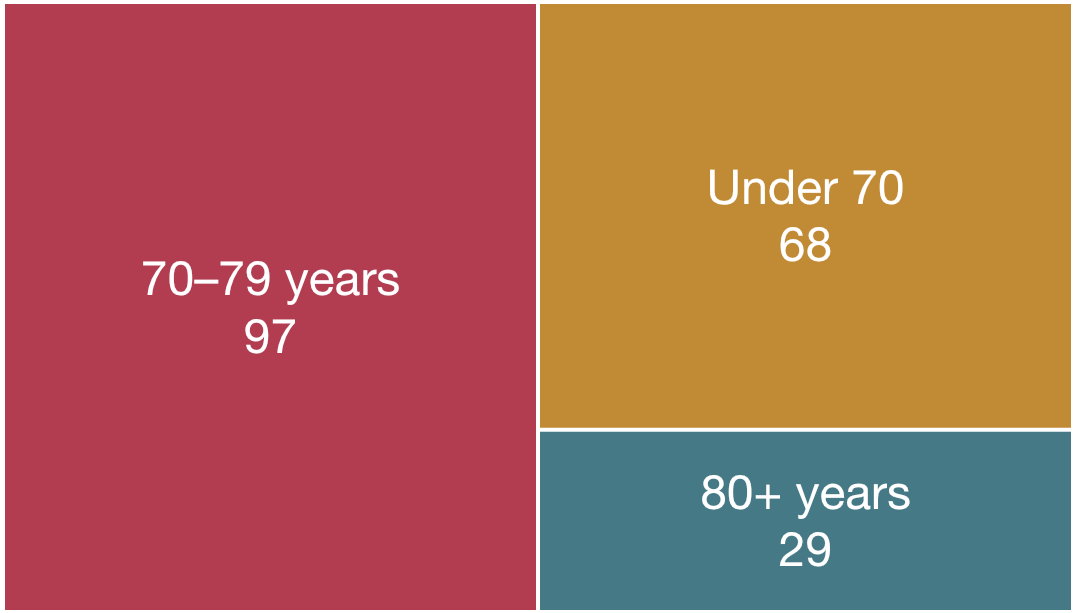
Pie chart

Shares of a total



Treemap

Parts sized by area



194 countries with valid life expectancy. Shares of countries, not people.

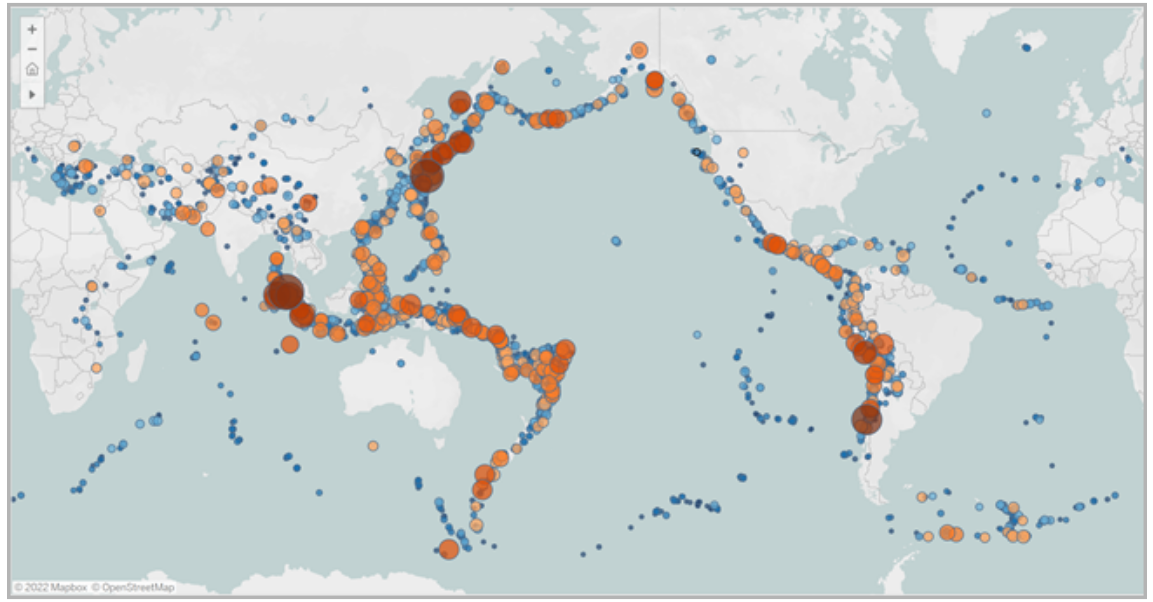
Life band → Color COUNTD(Country) → Angle / Size

Tableau chart types: geography

A map answers a question about location

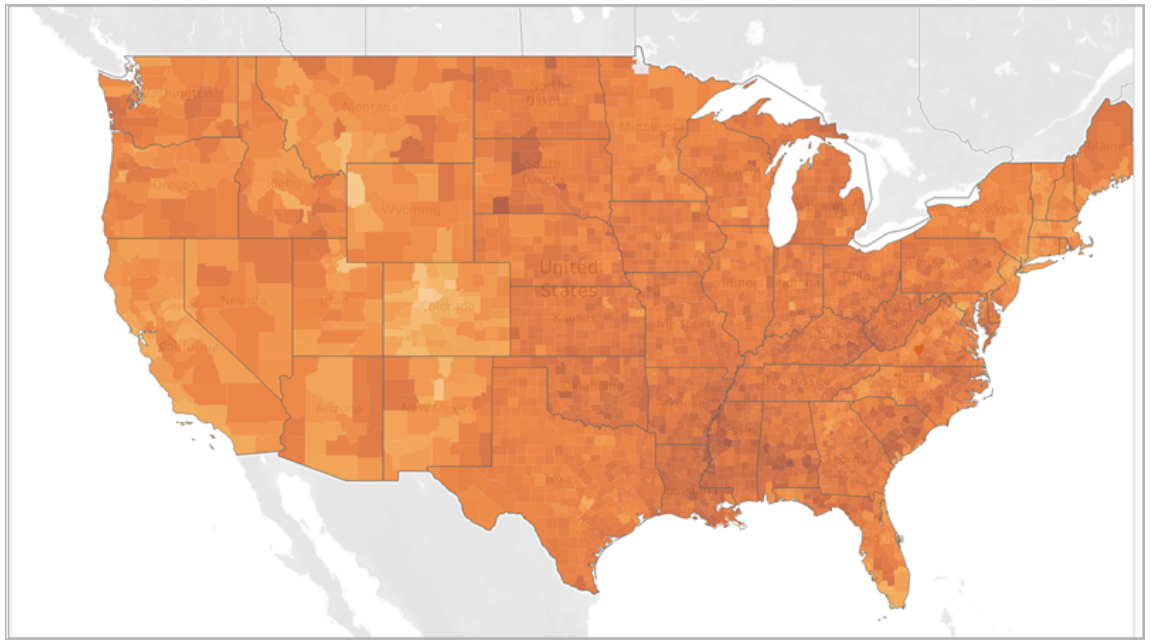
Symbol map

Size represents a value at each location.



Filled map

Color represents a value for each area.

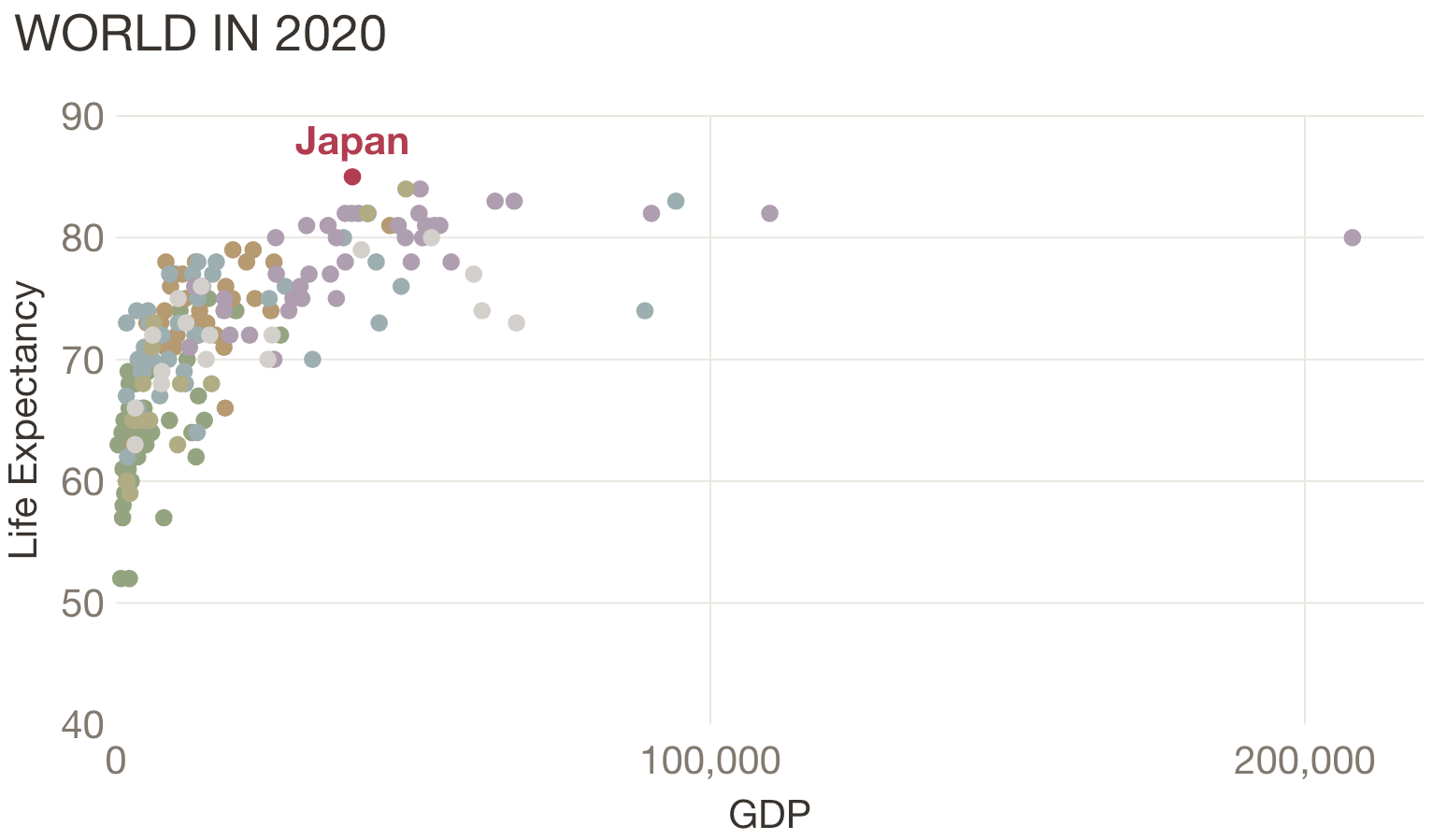


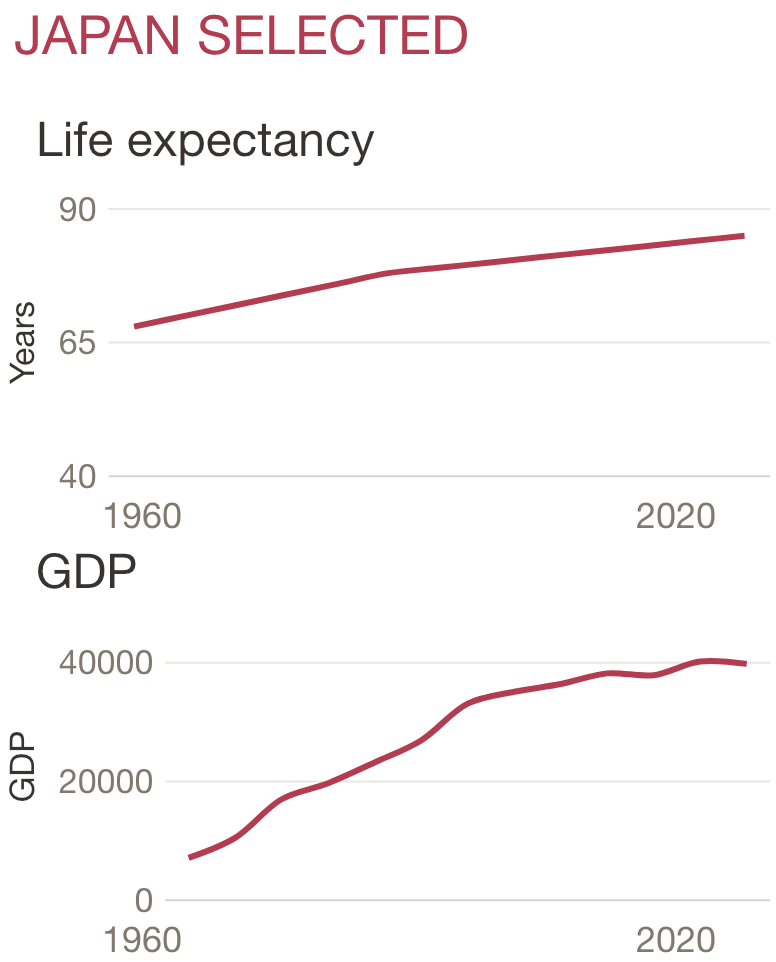
Try hans.csv: Country as a geographic field; Year = 2020; Life Expectancy on Color.

Map images: Tableau documentation examples

Today's tip: a Tableau dashboard

Last week's Gapminder chart, connected to a country's history





Click a country in the overview. Explore its history on the right.

The three worksheets

Right-click Year in the Data pane: Convert to Dimension

1 World

Columns	AVG(GDP)
Rows	AVG(Life Expectancy)
Filters	Year: 2020

Marks: Circle

Country on Detail
Region on Color

Filters: GDP > 0
Life Expectancy > 0

2 Life History

Columns	Year
Rows	AVG(Life Expectancy)
Filters	Year: 1960–2020

Marks: Line

Country on Color

Year pill menu: Continuous
Keep the full year range.

3 GDP History

Columns	Year
Rows	AVG(GDP)
Filters	Year: 1960–2020

Marks: Line

Country on Color

Duplicate Life History.
Replace the measure on Rows.

Measure pill menu: Measure > Average. Leave Pages empty for this dashboard.

The dashboard layout

A large overview and a smaller history column

1200 × 800



Tiled: horizontal container, with a vertical container on the right

1 Dashboard size

Turn Device Preview off. Select Default.
Dashboard pane > Size > Fixed size
Custom: Width 1200, Height 800

Drop sheets in the white canvas.
Gray space is outside the dashboard.

2 Tiled layout

World on the left.
Both histories in a vertical container on the right.

3 Year filter scope

Year filter menu: Apply to Worksheets
Selected Worksheets: World only
Histories keep 1960–2020.

Filtering with the sheet’s funnel icon

World sheet > Use as Filter

1 Select the sheet

Click World in the dashboard.

2 Use as Filter

Click the funnel icon on its toolbar.

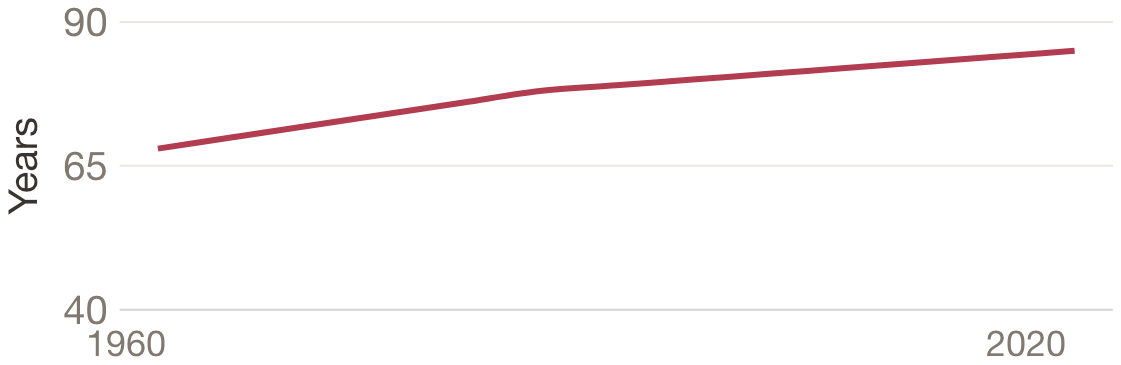
3 Select a country

Click Japan, then try India.

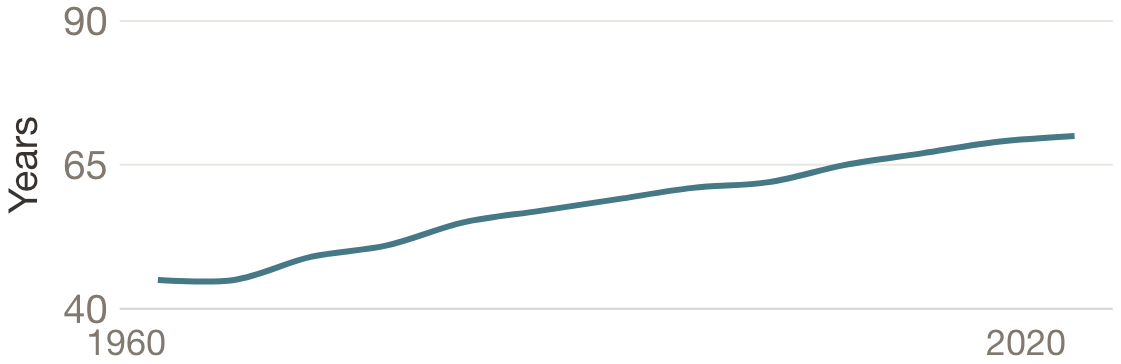
If a history collapses to 2020:

Dashboard > Actions: edit the generated filter.
Selected Fields: Country → Country only
Clear selection: Show all values

Click Japan



Then click India



Life expectancy, 1960–2020

Highlighting the selected country

Dashboard > Actions > Add Action > Highlight

Source sheet

World

Run action on

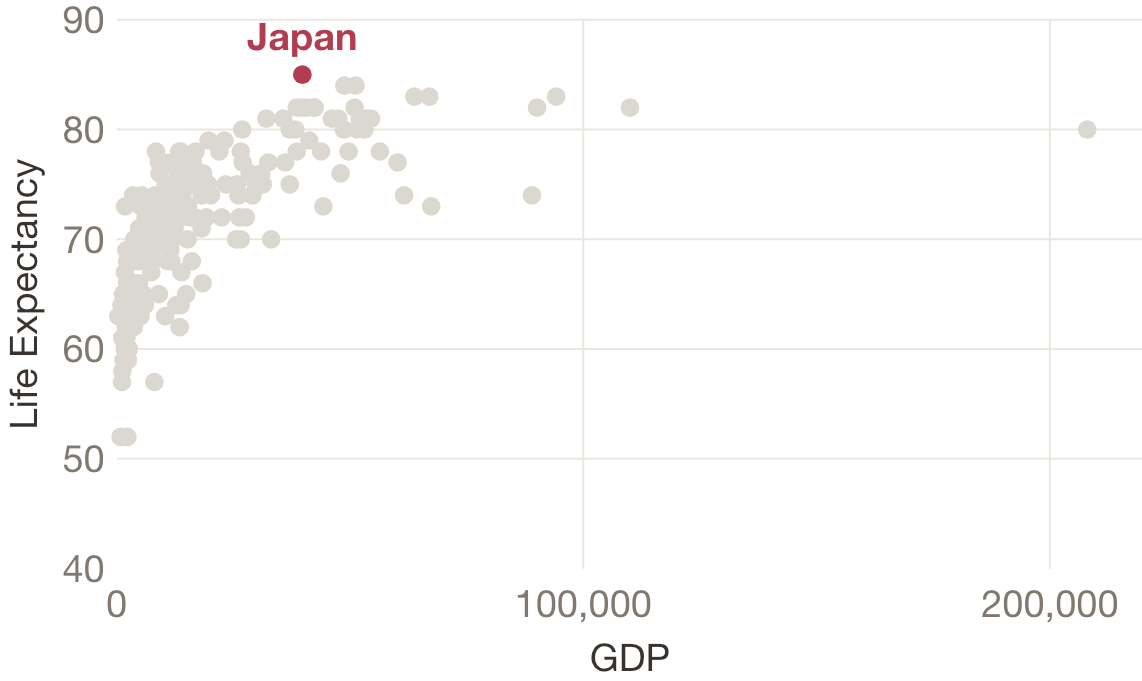
Select

Target sheet

World

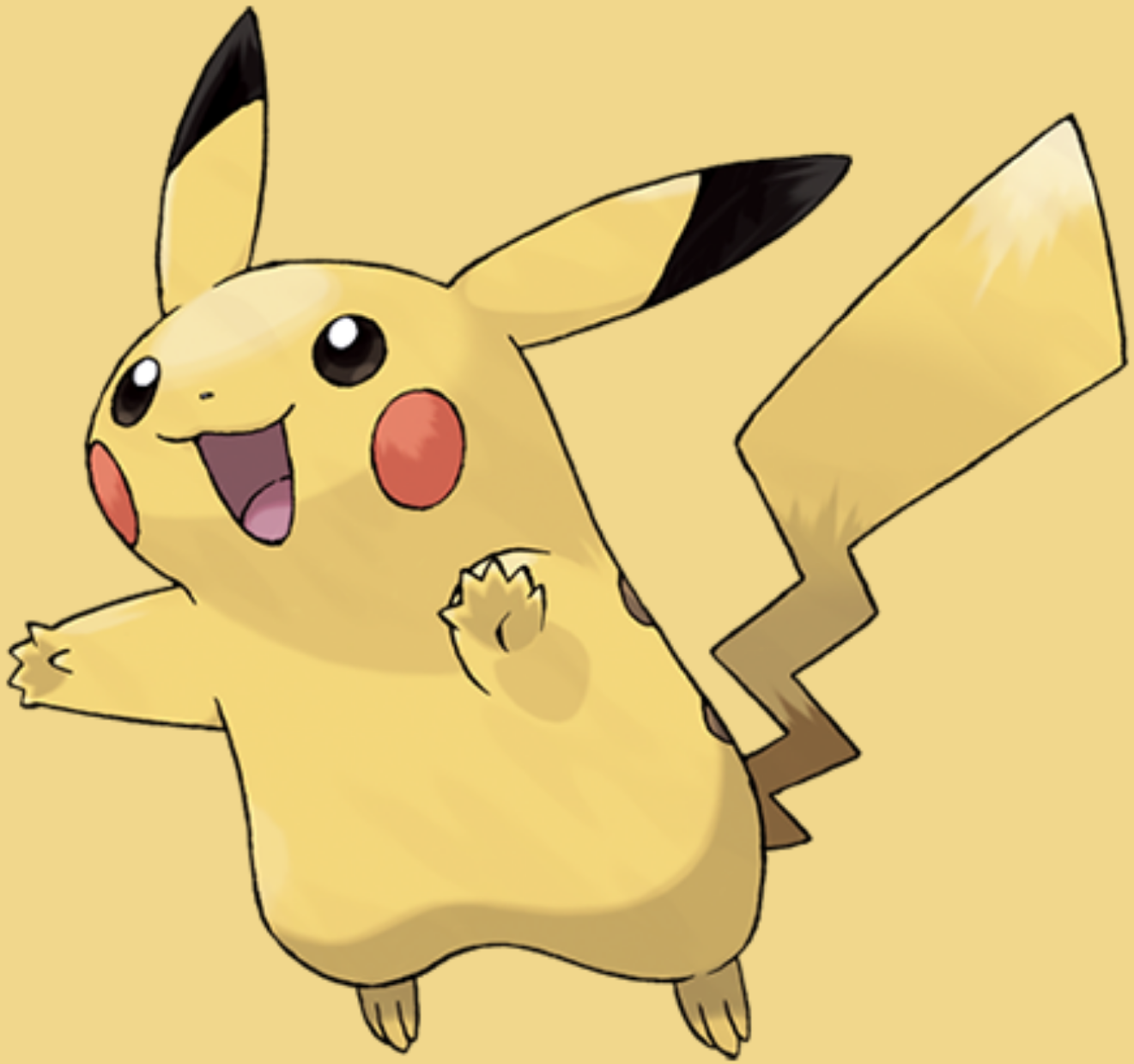
Target Highlighting

Selected Fields: Country



Click Japan: other circles fade.
Click empty space to clear.

Highlight: keep other countries visible. Filter: show only the selected country.



Pokémon Challenge

One dataset. Your question. No Pokémon knowledge needed.

15 min Solo

Choose a question. Make one chart in Tableau.
Give it a title that communicates your finding.

Groups of 4 Share

Everyone explains their finding and one visual choice.
Choose one chart for its insight and clarity.

One group submission

Submit names, title, finding and why you chose it.
Include one screenshot. A second is optional.

CSV + variable guide

Submit group work

Results / show and tell

A first chart with Pokémon images

Compare Attack for five Pokémon

01 Open the updated CSV

Text file: pokemon-review.csv. Open Sheet 1.

02 Choose five Pokémon

Name to Filters: Bulbasaur, Charmander, Eevee, Pikachu and Squirtle.

03 Enable image URLs

Right-click Image_URL in the Data pane. Choose Image Role > URL.

04 Build the rows

Rows: Image_URL first, then Name to its right.

05 Build the bars

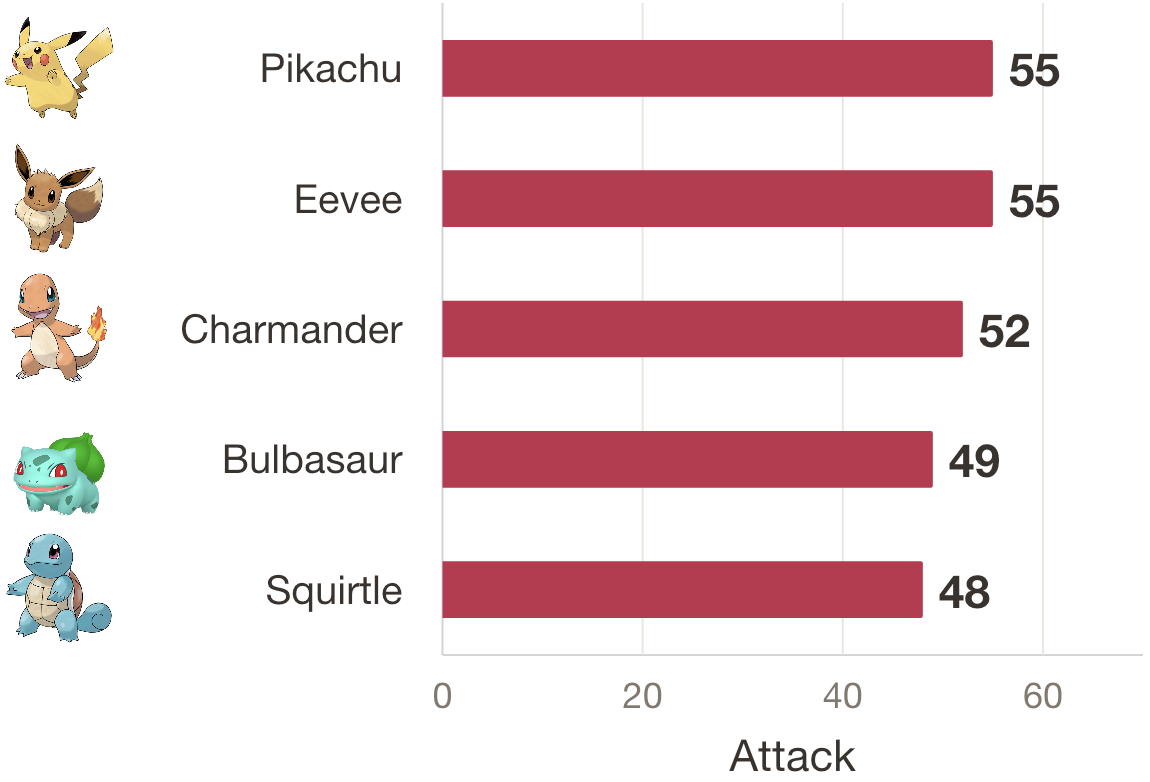
Attack to Columns. Pill menu: Measure > Average. On the Marks card, choose Bar.

06 Sort and label

Image_URL > Sort: Field, Attack, Average. Descending. Marks > Label: show mark labels.

07 Make the images larger

Widen the image column. Make rows about 80px tall. View: Standard. Attack axis starts at 0.



Images appear in row headers. Bars show average Attack.